Omega 117 (2023) 102841



Contents lists available at ScienceDirect

Omega

journal homepage: www.elsevier.com/locate/omega



# Efficient resilience portfolio design in the supply chain with consideration of preparedness and recovery investments☆



Riccardo Aldrighetti, Daria Battini, Dmitry Ivanov*

*University of Padova, Department of Management and Engineering, Stradella San Nicola 3, 36100 Vicenza, Italy*

ARTICLE INFO

*Article history:*
Received 2 February 2022
Accepted 13 January 2023
Available online 18 January 2023

*Keywords:*
Supply chain resilience
Disruption risk
Supply chain network design
Efficiency
Covid-19 pandemic

ABSTRACT

Supply chain (SC) resilience is imperative to cope with disruptions using some preparedness and recovery capabilities such as network redundancy (e.g., backup suppliers) and process flexibility (e.g., capacity agility). These capabilities frame an SC resilience portfolio. Both designing a resilient portfolio and recovering in case of a real disruption require investments. This paper presents a new mathematical model for designing an efficient resilience portfolio in a multi-echelon SC. Through computational and comparative analyses using a real-life case-study, we demonstrate that our model allows increasing resilience at minimal costs by determining an optimal combination of preparedness and recovery investments. Interestingly, the optimal solutions (i.e., efficient resilient SC designs) increase SC efficiency even in business-as-usual scenarios. This result contributes to the literature on transforming resilience from an expensive spend to a value-creation asset. We illustrate our approach using a real-life industrial example that allows for the identification of important relations between disruption duration/magnitude and efficiency of preparedness and recovery strategies. Based on computational, comparative, and case-study analyses, we deduce and generalize managerial implications at the network, supplier, and manufacturer levels. We take an extra step by extrapolating our major findings and generalized managerial implications toward the COVID-19 pandemic setting. The outcome of our research can be instructive for SC managers when deciding on investments in resilient redundancy allocation as a part of preparedness strategy and efficient recovery deployment.

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## 1. Introduction

Supply chain (SC) networks are exposed to disruptions that can cause adverse consequences such as negative financial effects, higher transportation costs, order delays, inventory shortages, and the loss of market shares [20,73]. These consequences can be mitigated by building resilient SCs [5,30,59]. SC resilience is imperative to cope with disruptions and comprises different preparedness assets and recovery activities that build an *SC resilience portfolio*.

In the context of SC disruption management, the term "portfolio" was first introduced by Sawik [69] to define "supply portfolio" as the allocation of demand for parts among selected suppliers. In [70], the term "resilient supply portfolio" was used to define the allocation of demand for parts among selected fortified and unfortified suppliers, along with selecting a subset of suppliers for protection against disruptions and pre-positioning of risk mitigation inventory among the protected suppliers. Sawik [71] enhanced the portfolio approach by simultaneous selection of primary and recovery supply portfolio to simultaneously optimize proactive and reactive sourcing decisions. Finally, a multi-portfolio approach was introduced in [72], where supply and demand portfolios are simultaneously optimized under disruptions of supply- and demand-side of SCs.

In this paper, we define SC resilience portfolio as an integrity of preparedness and recovery capabilities to mitigate disruptions and restore disrupted operations. The elements of a resilient SC portfolio can be classified into two groups. Proactive mitigation strategies are preparedness activities performed in anticipation of a disruption with the goal to increase the fortification and robustness of the SCs [52]. Reactive strategies aim to quickly adapt the SC and return it to its initial state after a real disruption [4,57,61,64]. We note that SC robustness is defined as "the ability to withstand a disruption (or a series of disruptions) to maintain the planned performance" [31], while SC resilience is "the firm's capability to withstand, adapt, and recover from disruptions to meet customer demand and ensure the target performance" [28].

Especially during the COVID-19 pandemic, companies understood the importance of resilience as a combination of redundancies and adaptation strategies required to survive through a crisis

☆ Area: Supply Chain Management. This manuscript was processed by Associate Editor Adenso-Diaz.

* Corresponding author: Supply Chain Management, Berlin School of Economics and Law, Badensche Str. 50, 10825 Berlin, Germany.
*E-mail address:* divanov@hwr-berlin.de (D. Ivanov).

https://doi.org/10.1016/j.omega.2023.102841
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[6,15,32,74]. For example, AGCO Corporation, an agricultural equipment manufacturer based in North America, applied various elements of resilience to tackle SC disruptions by adopting multiple sourcing and supplier diversification strategies. Before the pandemic, AGCO was already sourcing its materials both locally and from low-cost countries [32]. This combination of flexibility and robustness helped the company cope with the COVID-19 pandemic.

Both preparedness and recovery require some investments into a resilient SC portfolio. While resilience has been frequently considered as an expensive and cost-driving capability, recent studies call for the development of lean resilience concepts considering resilience from the efficiency and value-creation perspective [2,16]. For example, omni-channel distribution, diversified logistics networks, and strategic inventory locations can improve efficiency and resilience concurrently. Adding flexibility at the right place has been shown to be an efficient approach [68,77]. To support decision making on increasing SC resilience by robustness and flexibility, it is essential to develop sophisticated quantitative approaches [3,21,28,50]. However, the literature analysis shows that preparedness and recovery resilience capabilities have been mostly modelled independently from each other while their integration is crucial and desirable [52].

To close this research gap, we present a new decision-support model for optimizing the efficiency of the SC resilience portfolio in setting a multi-echelon, multi-period resilient SC design. We consider a resilience portfolio comprising a combination of several proactive and reactive resilience actions, i.e., backup suppliers, capacity recovery, extra capacity, facility fortification systems, multi-sourcing, and the (re)allocation of material flows. Our model accounts for disruption events that could impact both the supply and demand sides of the network. We propose an original formulation for decision-dependent scenario probability that allows considering a full spectrum of disruption scenarios with influence on actual activated sites (suppliers and facilities). The resultant endogenous stochastic nonconvex mixed-integer nonlinear model incorporates the risk aversion of decision makers. To the best of the authors' knowledge, this is the first study that considers a decision-dependent scenario probability formulation with the risk aversion of the decision maker in a resilient SC network design model with the optimization of resilience portfolio efficiency. We illustrate our model using a real-life industrial example that allows for the identification of important relations between disruption duration/magnitude and the efficiency of preparedness and recovery strategies. The outcomes of our research can be instructive for SC managers when deciding on investments in resilient redundancy allocation as a part of preparedness strategy and the efficient use of this redundancy at the recovery stages.

The organization of the paper is as follows. In Section 2, we review the related literature. Section 3 presents the mathematical formulation and main modeling features. A heuristic solution procedure is proposed and described. Successively, Section 4 presents the computational and comparative analysis of the model. The practical illustration is discussed in Section 5. Section 6 offers managerial insights. Section 7 concludes the study.

## 2. Literature review

Our paper builds on and contributes to two strands of the SC resilience literature: proactive mitigation plans and reactive contingency strategies; and the consideration of the risk-averse attitude of the decision maker.

### 2.1. Proactive mitigation plans and reactive contingency strategies

The literature has proposed different optimization models for designing resilient SC networks. For a general review of the literature, one can refer to Aldrighetti et al. [3], Dolgui and Ivanov [18], Ivanov and Dolgui [35], and Hägele et al. [24]. Robustness and resilience are two fundamental concepts for analyzing SC performance under disruptions [33]. Proactive and reactive strategies are two types of approaches to hedging against disruption and to increasing SC resilience [11]. The proactive approach builds robustness and accounts for possible perturbations without considering recovery actions [56,58,61]. Proactive mitigation strategies aim to plan for redundancies (e.g., buffer capacities, backup suppliers, prepositioned inventory, facility fortification) at the pre-disruption stage [33,47,85]. Backup suppliers are generally included to cope with supply-side disruptions, and they are frequently modelled as disruption-free sites that provide materials at a higher price [8,22,52,66]. Alternatively, adding extra capacity to suppliers/facilities can help increase robustness and compensate for capacity losses caused by disruption events [22,41,66,73,89]. Regarding facility protection, two main methods to model this resilience strategy are as follows: (i) by allowing complete reliability of some sites (i.e., disruption immunity) or (ii) by considering a fortification system that reduces the disruption impact and capacity losses. Abbasi et al. [1] considered how hubs and distribution centres could be fortified, making them immune to failures. Fattahi et al. [19] and Gholami-Zanjani et al. [22] developed the possibility to invest in fortification levels to reduce warehouses' disrupted capacity.

Reactive strategies aim to adjust SC processes and structures after a disruption occurs. This can be done through parametrical or structural adaptions according to the severity of the disruption [4]. Most of the models in the field of resilient SC design permit the operational reallocation of material flows based on the remaining available assets [9,25,84]. Another approach is to preplan for a primary allocation flow and a backup one that is activated only in the event of a disruption occurring on the primary site [1]. In addition, Fattahi et al. [19] and Sabouhi et al. [66] considered transshipping the materials from disrupted sites to available ones to rebalance the system inventory and improve customer service with a limited increase in costs. Other works [52,89] considered the possibility of recovering the lost capacity so as to restore the initial SC structure.

### 2.2. Risk aversion of the decision maker

Location models for planning reliable systems can be grouped into two main categories: risk-neutral and risk-averse models [48,75]. The vast majority of research has focused on risk-neutral decision models when facilities are assumed to fail based on predefined probabilities [73,80]. However, disruptions, by definition, are rare events and are thus difficult to predict [33,79]. It may be difficult to obtain data, and subjective probabilities must often be used. On the other hand, risk-averse models allow the decision maker to quantify, control, and limit the risk based on his or her preferences.

Risk aversion can be incorporated in different ways. We now discuss different risk-averse measures used to mitigate the impact of SC disruptions. First, robust optimization incorporates variance as a measure of variability into the objective function. In this context, the decision-maker attitude is included through a parameter that limits the tradeoff between the expected value and maximum regret [9,40]. Second, a practical use of CVaR (conditional value at risk) in SC disruption management is shown in [69], which leads to linear stochastic programs should be emphasized as a highly efficient computational approach. In the CVaR approach, selection of confidence level allows the decision maker to focus on specific percentage of worst-case scenario outcomes. As a result, low-probability, high-impact and high-probability, low-impact scenarios can be considered simultaneously and the issue of "unbalanced probability" can be eliminated. Other methods are optimiz-

ing worst-case scenarios [73,75], including risk measures such as $p$-robustness [25,84], the VAR (value at risk) [51], the resilience index [53], and (semi-)deviations from targets [14,19].

In particular, scenario-based stochastic programming is a widely employed modeling framework in the field of resilient SC to handle uncertain parameters in optimization models. Uncertain parameters can be classified into two categories: exogenous uncertainty and endogenous uncertainty [26]. In an exogenous stochastic program, decisions are made, and uncertain values are revealed at distinct points in time. Some decisions are made before the actual values of the uncertain parameters are known, but the realization of the stochastic parameters is independent of the decisions. Therefore, exogenous uncertainty problems are problems where the distribution of the stochastic parameters is known; for example, it could be based on historical data or expert opinion. On the other hand, endogenous uncertainty problems are problems where decisions at one point in time will have a substantial impact on the uncertainty faced later, either in terms of when information about the actual value of a stochastic parameter becomes available or the probability that a certain realization of a parameter occurs. A problem is classified as having endogenous uncertainty when the decisions associated with the problem to be solved influence the uncertainty of parameters that are a part of the problem. The endogenous decisions are considered decision-dependent uncertainty.

### 2.3. Theoretical positioning and main contributions

Our analysis shows that previous studies focused on facility location and allocation under exogenous stochastic parameters [26]. Only a few studies [89] developed an endogenous stochastic resilient SC design where the remaining production capacity available after disruption occurrence depends on first-stage decision variables. Most of the existing models do not consider the specific recovery activities of a disrupted SC network and focus on preparedness that maintains SC functionality after a disruption event. In other words, recovery strategies are rarely addressed in comparison to mitigation strategies. Finally, the literature regarding risk-averse resilient SC network design models with disruption consideration is still scarce [13,80].

Considering this research gap, our paper proposes a novel approach to optimize the efficiency of the resilience portfolio while designing SCs with the consideration of both preparedness for the disruption events and recovery. We formulate a two-stage endogenous stochastic nonconvex mixed-integer nonlinear model for a multi-period capacitated three-echelon SC network design. This allows for determining the optimal facility number, locations, capacity, and allocation with disruption consideration on both the supply and demand sides of the network. With the help of our approach, it becomes possible to design an optimal portfolio of proactive and reactive resilience strategies to provide a deeper understanding of how to plan for redundancies and recovery actions to enhance SC resilience with alternative disruption scenarios. This greatly informs decisions regarding a tradeoff between the effectiveness of adopting resilience strategies and the efficiency of redundancy and recovery actions.

In summary, we contribute to the literature in two distinct areas. First, our research simultaneously considers proactive mitigation plans and contingency strategies in the resilience portfolio in the real-life setting of multi-echelon networks. Second, our decision-support model incorporates risk aversion of the decision maker in two ways: (i) a decision-dependent scenario probability formulation that enables consideration for a full spectrum of disruption scenarios with influence on actual activated sites (suppliers and facilities); and (ii) the decision maker's ability to increase his/her risk-averse attitude by varying the number of possible events that can happen in a scenario. In their totality, these contributions advance the existing applications of operations research to resilient SC design with a comprehensive consideration of a practical tradeoff between efficiency and resilience. For a detailed comparison about modeling the characteristics of our study among some remarkable works in the topic published in the last few years, the reader is referred to Table A1 in the appendix.

## 3. Problem statement and mathematical model

In this section, we present our formulation of a three-echelon SC network design consisting of suppliers, facilities (which can be both production and distribution centers), and customer zones. Fig. 1 illustrates an example of the SC design structure and resilience portfolio.

The SC is modelled subject to the following characteristics:

- There are two types of facilities: already existing facilities and new (potential) facilities.
- At the existing facilities, the model can decide to increase the already installed capacity by adding some extra capacities and to invest in a protection system.
- For new (potential) facilities, among the available locations, the model decides the best locations and the associated capacity.
- There are multiple periods involved with disruptions and recovery decisions.
- A single type of product is considered.

Primary suppliers are capacitated based on predefined contracted capacity levels $QS_l$. Facilities can procure the materials from a multitude of suppliers (i.e., no single-sourcing constraint). Facilities are capacitated based on some predefined capacity levels $QF_l$. The model permits the inclusion of already established facilities, thereby considering possible reconfigurations of the existing SC networks. It is assumed that customer zones are predetermined and fixed, and each customer zone can be supplied by multiple facilities. The unsatisfied demand of each customer zone is subject to penalty costs. To mitigate disruption effects and increase SC resilience, the model allows for increasing the contracted capacity from suppliers and/or the installed capacity at facilities. Customers could also be reassigned based on facility availabilities through flow reallocation. Three resilient specific features are included:

- *Backup suppliers*: Facilities can decide to establish a backup contract with suppliers. Backup suppliers are considered to be reliable sites where the SC needs to pay an initial fee for establishing the contract, and the supplies are subject to a penalty cost because of its source (only periodic and urgent). In general, backup suppliers are more expensive and economically advantageous for small quantities of supplies [17,82].
- *Protection systems*: Facilities can be reinforced through investment in protection systems. In this context, flexibility can be achieved by increasing component commonalities, postponements, and risk-pooling strategies or by setting up more flexible production systems and Industry 4.0 technologies [16]. This resilience investment is expressed as a percentage of the standard facility establishment costs through a set $P$ of protection levels $p$. Each level allows a reduction of $\beta_p$ in the disruption magnitude, and the new disrupted capacity will be $\theta_i \cdot \beta_p$, where $\theta_i$ is a disrupted capacity at facility $i$ as a percentage of nominal capacity. Facilities cannot become completely reliable.
- *Planning recovery activities to restore the lost capacity*: Disruptions are characterized by a duration *DUR* of periods. At the end of a period, facilities have the possibility to invest in the recovery of the lost capacity until they reach their nominal capacity as installed at period $t = 0$.

Recovery actions are permitted only at facilities as suppliers are, in general, external entities and could decide to recover or not. In

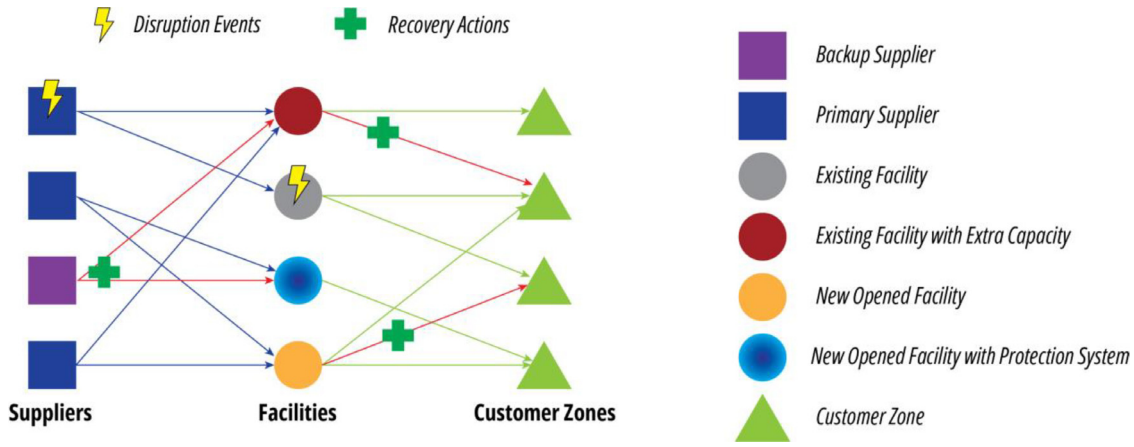



**Fig. 1.** Supply chain design structure and resilience portfolio.

addition, we do not consider recovery actions to be mandatory—i.e., a disrupted facility does not need to (fully) restore its production/distribution capacity after a disruption. This feature is motivated by the fact that the full recovery of a disrupted facility is not practical given that the post-disruption period could result in resource limitation [52]. A disruption may cause substantial changes in demand locations and type. SC managers could decide to recover or not recover the lost assets based on the needs after a disruption. For example, in the COVID-19 pandemic setting, the return to the initial performance and operational plans became complicated because of radically changed demand and supply dynamics [30,34,34,65,86]. Periodic-based scenario modeling is adopted to include the dynamics of different periods along with some specific scenarios, thereby accounting for disruption risks. All primary suppliers and facilities are subject to possible disruption events that may reduce the installed/contracted capacity by percentage $\theta_k$ and $\theta_i$, respectively. Six different parameters are modelled to describe the evolution of capacity in supplier and facilities:

- $QN_k$, $QN_i$: nominal capacity setup at the first period for suppliers and facilities, respectively.
- $QP_i$: protected capacity reliable on disruption effects through the flexibility/protection system.
- $Q^s_{kt}$, $Q^s_{it}$: current available capacity in each period for suppliers and facilities, respectively.
- $QD^s_{kt}$, $QD^s_{it}$: disrupted capacity that reduces current available capacity for suppliers and facilities.
- $QR^s_{it}$: capacity that is feasible to be recovered after disruption occurrence (only at facilities).
- $RQ^s_{it}$: actual recovered capacity in a period.

Other assumptions are as follows:

- There is no disruption at any site at period $t = 0$.
- A disruption at period $t$ decreases the capacity in the same period $t$.
- A recovery action planned at period $t$ will be fulfilled in the next period. Therefore, recovery costs are registered at period $t$, but the capacity will become available in period $t + 1$.
- The last available period for making recovery actions is period $T - 1$.

The goal of the proposed model is fourfold: (i) to determine the optimal primary suppliers and facility locations, capacity, and number; (ii) to plan optimal proactive resilience investment in terms of backup suppliers and facility protection systems; (iii) to allocate primary and backup upstream and downstream flows; and (iv) to design the best recovery actions for reacting to disruptions. A two-stage mixed-integer nonlinear model is derived in Section 3.1. The problem notations are detailed in Table 1.

### 3.1. Model formulation

$$\begin{aligned}\min\ TC = &\sum_k \left[\sum_l (CPS + CPQ_k \cdot QS_l)\cdot f_{kl}\right] \\ &+ \sum_k [CBS_k \cdot b_k] + \sum_{i\in NF}\sum_l (CL_i + CC_i \cdot QF_l)\cdot y_{il} \\ &+ \sum_{i\in OF}\sum_l (CC_i \cdot QF_l \cdot z_{il}) + \sum_l \sum_l QF_l \cdot CC_i \cdot \sum_p \left(\varepsilon_p \cdot w_{ilp}\right) \\ &+ \sum_S P_s \sum_t \left\{\sum_i \left[\sum_k \left(CP_k + CT\cdot \overline{D}_{ki}\right)\cdot x^s_{kit} \right.\right. \\ &+ \sum_k \left(CP_k + CT\cdot \overline{D}_{ki}\right)\cdot \gamma \cdot u^s_{kit} + \sum_j \left(CT\cdot \overline{D}_{ij} + CH\right)\cdot x^s_{ijt} \\ &\left.\left. + \left(CC_i \cdot \left(1 - \frac{t}{T}\right) + CH\right)\cdot QD^s_{it} + CC_i \cdot RQ^s_{it}\right] + \sum_j \varphi \cdot u^s_{jt}\right\}\end{aligned} \tag{1}$$

The objective function (1) seeks to minimize the total expected cost. The first two rows are related to investment costs, while operational costs are represented in the third and fourth rows of the expression. Consistent with Jabbarzadeh et al. [40], Sabouhi et al. [66], and Gholami-Zanjani et al. [22], we included the investment costs for contracting with primary suppliers and activating a relationship with backup suppliers [22,52,66], the cost of building new facilities and increasing the capacity of already established facilities, and the possibility to provide a protection/flexibility system to both new and already established facilities [19,40]. In terms of operational costs, we consider the purchasing and transportation of supplies from primary contracted and backup suppliers [22,40,66], the transportation cost for delivering products to the customer zone, the holding cost at facilities, the damage costs caused by disruption occurrence, and the recovery cost for restoring lost capacity [19]. The last term of the expression describes the backlog cost.

$$QN_k = \sum_l QS_l f_{kl} \forall k \in K \tag{2}$$

$$QD^s_{kt} = 0 \forall k \in K,\ t = 0,\ \forall s \in S \tag{3}$$

**Table 1**
Notations.

| *Notation* | *Definition* |
|---|---|
| *Set and indices* | |
| $K$ | set of suppliers, indexed by $k$ |
| $OF$ | set of already opened facilities |
| $NF$ | set of new possible location for facilities |
| $I$ | set of facilities, indexed by $i$ ($I$ = OFNF) |
| $J$ | set of customers, indexed by $j$ |
| $L$ | set of capacity levels, indexed by $l$ |
| $P$ | set of protection levels, indexed by $p$ |
| $T$ | set of time periods, indexed by $t$ |
| $S$ | set of scenarios, indexed by $s$ |
| *Parameters* | |
| $A^s_{it}$ | 1 if facility $i$ is disrupted at period $t$ in scenario $s$; 0 otherwise |
| $A^s_{kt}$ | 1 if supplier $k$ is disrupted at period $t$ in scenario $s$; 0 otherwise |
| $\beta_p$ | percentage reduction of disrupted capacity after investing in protection systems at level $p$ |
| $CBS_k$ | cost of establishing a backup contract with supplier $k$ |
| $CC_i$ | unitary cost of capacity for facility $i$ |
| CH | holding costs |
| $CI_i$ | fixed cost of opening facility $i$ |
| CPS | fixed cost of contracting with a primary supplier |
| $CPQ_k$ | unitary cost of contracted capacity with supplier $k$ |
| $CP_k$ | unitary purchasing cost from supplier $k$ |
| CT | transportation costs per unit of product and distance |
| $\bar{D}_{ij}$ | distance between facility $i$ and customer $j$ |
| $\bar{D}_{ki}$ | distance between supplier $k$ and facility $i$ |
| $D_{jt}$ | periodic demand of customer $j$ |
| DUR | disruption duration |
| $\varepsilon_p$ | portion of capacity that is protected through a flexibility/protection system at protection level $p$ |
| $\gamma$ | penalty applied for purchasing from backup suppliers |
| M | a big number |
| N | maximum number of disruption events that can happen in a scenario |
| $N_D_s$ | number of disruption events modelled in scenario $s$ |
| $N_AD_s$ | number of disruption events that impact on an actual opened facility and/or selected primary supplier in scenario $s$ |
| $OPENCAP_i$ | available capacity at already opened facilities $i$ |
| $QF_l$ | facility's capacity at level $l$ |
| $QS_l$ | primary supplier's contracted capacity at level $l$ |
| T_DIS | period in which disruption event occurs |
| $\varphi$ | penalty cost applied for missing unit of demand |
| $P_s$ | probability associated with scenario $s$ |
| $\theta_i$ | disrupted capacity at facility $i$ as a percentage of nominal capacity |
| $\theta_k$ | disrupted capacity at supplier $k$ as a percentage of contracted capacity |
| $WEIGHT_s$ | weight of scenario $s$ for probability calculation |
| *Decision variables* | |
| $f_{kl}$ | 1 if supplier $k$ is selected as primary supplier with contracted capacity at level $l$; 0 otherwise |
| $b_k$ | 1 if supplier $k$ is selected as backup supplier; 0 otherwise |
| $y_{il}$ | 1 if facility $i$ is opened with capacity level $l$; 0 otherwise |
| $z_{il}$ | 1 if facility $i$'s capacity is increased with capacity level $l$; 0 otherwise |
| $w_{ilp}$ | 1 if facility $i$ opened with capacity level $l$ is reinforced with protection level $p$; 0 otherwise |
| $pfa_i$ | 1 if protection/flexibility investment is planned for already opened facility i; 0 otherwise |
| $r^s_{ilt}$ | 1 if capacity level $l$ is recovered at facility $i$ at period $t$ in scenario $s$; 0 otherwise |
| $x^s_{kit}$ | product flows between supplier $k$ and facility $i$ at period $t$ in scenario $s$ |
| $x^s_{ijt}$ | product flows between facility $i$ and customer zone $j$ at period $t$ in scenario $s$ |
| $u^s_{kit}$ | backup flows between backup supplier $k$ and facility $i$ at period $t$ in scenario $s$ |
| $u^s_{jt}$ | lost demand of customer zone $j$ at period $t$ in scenario $s$ |
| *Variables* | |
| $QN_i$ | nominal capacity installed at facility $i$ |
| $QP_i$ | protected capacity reliable on disruption effects through flexibility/protection system at facility $i$ |
| $ProtCap_i$ | auxiliary variable to represent the total capacity on which to apply protection system at already opened facility $i \in OF$ |
| $Q^s_{it}$ | available capacity at facility $i$ at period $t$ in scenario $s$ |
| $QD^s_{it}$ | disrupted capacity at facility $i$ at period $t$ in scenario $s$ |
| $QR^s_{it}$ | recoverable capacity at facility $i$ at period $t$ in scenario $s$ |
| $RQ^s_{it}$ | capacity recovered at facility $i$ at period $t$ in scenario $s$ |
| $QN_k$ | contracted capacity with primary supplier $k$ |
| $Q^s_{kt}$ | available capacity at supplier $k$ at period $t$ in scenario $s$ |
| $QD^s_{kt}$ | disrupted capacity at supplier $k$ at period $t$ in scenario $s$ |

$$Q^s_{kt} = QN_k \forall k \in K,\ t = 0,\ \forall s \in S \tag{4}$$

$$QD^s_{kt} = Q^s_{k\ t-1}\theta_k A^s_{kt} \forall k \in K,\ \forall t \in T,\ t > 0, \forall s \in S \tag{5}$$

$$Q^s_{kt} = Q^s_{k\ t-1} - QD^s_{kt} \forall k \in K,\ \forall t \in T,\ t > 0, \forall s \in S \tag{6}$$

$$QN_i = OPENCAP_i + \sum_l QF_l \cdot z_{il} \forall i \in OF \tag{7}$$

$$QN_i = \sum_l QF_l y_{il} \forall i \in NF \tag{8}$$

$$QP_i = \sum_l QF_l \cdot \sum_p w_{ilp}\beta_p \forall i \in I \tag{9}$$

$$QD^s_{it} = 0 \forall i \in I,\ t = 0,\ \forall s \in S \tag{10}$$

$$QR^s_{it} = 0 \forall i \in I,\ t = 0 t = T,\ \forall s \in S \tag{11}$$

$$Q_{it}^{s} = QN_i \forall i \in I,\ t = 0,\ \forall s \in S \tag{12}$$

$$QD_{it}^{s} = \left(Q_{i\ t-1}^{s} - QP_i\right)\theta_i A_{it}^{s} \forall i \in I,\ \forall t \in T,\ t > 0,\ \forall s \in S \tag{13}$$

$$QR_{it}^{s} = QN_i - Q_{it}^{s} \forall i \in I,\ \forall t \in T,\ t > (T_DIS_i + DUR),\ \forall s \in S \tag{14}$$

$$RQ_{it}^{s} = \sum_{l} QF_l \cdot \theta_i \cdot r_{ilt}^{s} \forall i \in I,\ \forall l \in L,\ \forall t \in T,\ t > 0,\ \forall s \in S \tag{15}$$

$$Q_{it}^{s} = Q_{i\ t-1}^{s} - QD_{it}^{s} + RQ_{i\ t-1}^{s} \forall i \in I,\ \forall t \in T,\ t > 0,\ \forall s \in S \tag{16}$$

$$RQ_{it}^{s} \leq QR_{it}^{s} \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{17}$$

Eqs. (2)–(6) and (7)–(17) are the capacity definitions for suppliers and facilities, respectively. Eq. (2) defines the nominal contracted capacity with primary suppliers. Eqs. (3) and (4) represent the disrupted and available capacity of the primary suppliers at the first period ($t = 0$) at each scenario, while Eqs. (5) and (6) are the capacity definitions for the rest of the periods. For facilities, Eqs. (7) and (8) define the nominal capacity at new and already established facilities, respectively, while Eq. (9) indicates the protected capacity as a result of investment in the protection system. The disrupted, recoverable, and available capacities for the first period are defined through Eqs. (10)–(12). For the rest of the periods, Eqs. (13)–(16) are related to the disrupted, recoverable, recovered, and available capacities, respectively. Constraints (17) assure that the recovered capacity does not exceed the maximum recoverable capacity.

$$\sum_{l} f_{kl} + b_k \leq 1 \forall k \in K \tag{18}$$

$$\sum_{i} x_{kit}^{s} \leq Q_{kt}^{s} \forall k \in K,\ \forall t \in T,\ \forall s \in S \tag{19}$$

$$\sum_{s}\sum_{t}\sum_{i} u_{kit}^{s} \leq M \cdot b_k \ \forall k \in K \tag{20}$$

Constraints (18) ensure that if a supplier relationship is established, it could be set as a primary supplier agreement with only one capacity level or as a backup supplier. Constraints (19) ensure that the total outer flow of a supplier does not exceed the total available capacity. Constraints (20) guarantee that the backup supply flow is allowed only from established backup suppliers.

$$\sum_{l} y_{il} \leq 1 \forall i \in NF \tag{21}$$

$$\sum_{l} z_{il} \leq 1 \forall i \in OF \tag{22}$$

$$OPENCAP_i + \sum_{l} QF_l \cdot z_{il} \leq QF_L \forall i \in OF \tag{23}$$

$$\sum_{l}\sum_{p} w_{ilp} \leq 1 \forall i \in I \tag{24}$$

$$\sum_{l} QF_l \cdot \sum_{p} w_{ilp} \leq \left(OPENCAP_i + \sum_{l} QF_l \cdot z_{il}\right) \cdot pfa_i \forall i \in OF \tag{25}$$

$$QF_l \cdot \sum_{p} w_{ilp} \leq QF_l \cdot y_{il} \forall i \in NF,\ \forall l \in L \tag{26}$$

$$\sum_{j} x_{ijt}^{s} \leq Q_{it}^{s} \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{27}$$

$$\sum_{i} x_{ijt}^{s} + u_{jt}^{s} = D_j \forall j \in J,\ \forall t \in T,\ \forall s \in S \tag{28}$$

$$\sum_{k} \left(x_{kit}^{s} + u_{kit}^{s}\right) \geq \sum_{j} x_{ijt}^{s} \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{29}$$

$$\sum_{l} r_{ilt}^{s} \leq 1 \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{30}$$

Eqs. (21)–(22) assure that only one level of capacity could be selected for new facilities or as additional capacity to already existing facilities. Eq. (23) guarantees that the total installed capacity of already existing facilities does not exceed the maximum capacity level. Eq. (24) ensures that facilities can be protected at only one level of protection, while Eqs. (25) and (26) are necessary for applying the protection investment to the right capacity level for already established and new facilities, respectively. Eq. (27) restricts the facility flow to the available capacity, while Eq. (28) specifies the demand needs to be fully served by facilities; otherwise, penalty costs occur. Eq. (29) represents the balance constraint between inbound and outbound flows. Eq. (30) guarantees that only one level of capacity can be recovered in each period.

$$f_{kl} \in \{0,1\} \forall k \in K,\ \forall l \in L \tag{31}$$

$$b_k \in \{0,1\} \forall k \in K \tag{32}$$

$$y_{il} \in \{0,1\} \forall i \in NF,\ \forall l \in L \tag{33}$$

$$z_{il} \in \{0,1\} \forall i \in OF,\ \forall l \in L \tag{34}$$

$$w_{ilp} \in \{0,1\} \forall i \in I,\ \forall l \in L,\ \forall p \in P \tag{35}$$

$$pfa_i \in \{0,1\} \forall i \in OF \tag{36}$$

$$r_{ilt}^{s} \in \{0,1\} \forall i \in I,\ \forall j \in J,\ \forall t \in T,\ \forall s \in S \tag{37}$$

$$x_{kit}^{s} \in \mathbb{N} \forall k \in K,\ \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{38}$$

$$u_{kit}^{s} \in \mathbb{N} \forall k \in K,\ \forall i \in I,\ \forall t \in T,\ \forall s \in S \tag{39}$$

$$x_{ijt}^{s} \in \mathbb{N} \forall i \in I,\ \forall j \in J,\ \forall t \in T,\ \forall s \in S \tag{40}$$

$$u_{jt}^{s} \in \mathbb{N} \forall j \in J,\ \forall t \in T,\ \forall s \in S \tag{41}$$

Finally, Eqs. (31)–(41) define the nature of the decision variables.

$$protCap_i \in \mathbb{N} \forall i \in OF \tag{42}$$

$$\sum_{l} QF_l \cdot \sum_{p} w_{ilp} = ProtCap_i \forall i \in OF \tag{43}$$

$$ProtCap_i \geq 0 \forall i \in OF \tag{44}$$

$$ProtCap_i \leq QF_L \cdot pfa_i \forall i \in OF \tag{45}$$

$$ProtCap_i \leq \left(OPENCAP_i + \sum_{l} QF_l \cdot z_{il}\right) \forall i \in OF \tag{46}$$

$$ProtCap_i \geq \left(OPENCAP_i + \sum_{l} QF_l \cdot z_{il}\right) - QF_L \cdot (1 - pfa_i) \forall i \in OF \tag{47}$$

Constraints (25) are nonlinear given the product between two decision variables. To linearize, a new integer variable $protCap_i$ was defined in (42) such that constraints (25) become (43) together with the additional constraints (44)–(47).

### 3.2. Scenario generation

Research in stochastic programming context has extensively dealt with generating efficient scenarios [27,42]. Scenario-based formulations may lead to a solution that is intractable given the significant number of scenarios involved. Most of the studies generally exploited a small predetermined set of scenarios with fixed probabilities for their stochastic programs [3]. In our model, we propose a decision-dependent uncertainty set to describe the scenario generation procedure that can reduce the number of tested scenarios, along with adopting a risk-averse attitude for the decision maker.

#### 3.2.1. Initial sample definition

*ScGen1* The nominal scenario with no disruption occurrence is excluded. Business-as-usual situations are, by definition, much more likely than disrupted scenarios. The inclusion of a mix of business-as-usual and disruption situations with assigned probability might lead to unbalanced probabilities with significantly lower values for disrupted scenarios. This might yield optimal solutions based on a disruption-free setting. Therefore, to maintain the risk aversion of the decision maker, each generated scenario needs to consider at least one disruption event.

*ScGen2* The maximum number of disruption events that can happen in a scenario is defined through the parameter $N$. Then the total theoretical scenario size is equal to the combination explained in equation (48).

$$|S| = \sum_{n=1}^{N} \binom{|K| + |I|}{n} \tag{48}$$

#### 3.2.2. Scenario reduction

*ScRed1* The risk aversion of the decision maker is controlled through $N$. The bigger the value of $N$, the higher the risk-aversion attitude. To avoid a problem size that cannot be controlled, considering that disruption risk is associated with rare events, we assume that at most, two disruption events can happen per scenario ($N = 2$), as echoed by such researchers as Hatefi and Jolai [25] and Sanci et al. [67].

*ScRed2* Scenario reduction is performed through the definition of a scenario probability that depends on the first-stage decision. A scenario is included in the final set if the considered disruption event(s) (modelled through the $A^s_{kt}$ and $A^s_{it}$ parameters) impact the operations of actual opened facilities or selected primary suppliers. The parameter $N_D_s$ [Eq. (49)] represents the number of disruption events modelled in a scenario $s$. $N_AD_s$ [Eq. (50)] represents the number of disruption events that impact on an actual opened facility and/or selected primary supplier. The scenario weight is defined in Eq. (51), where scenarios with fewer disruptions have higher weight compared to scenarios with more disruption events and that are less probable and therefore have smaller weights. Thanks to Eq. (51), scenarios that consider disruption events at non-opened facilities and/or non-selected suppliers (i.e., $N_AD_s \neq N_D_s$) are associated with a weight of 0 and thus excluded from the considered set.

$$N_D_s = \sum_t \left[ \sum_k A^s_{kt} + \sum_i A^s_{it} \right] \tag{49}$$

$$N_AD_s = \sum_t \left[ \sum_k \left( A^s_{kt} \cdot \sum_l f_{kl} \right) + \sum_i \left( A^s_{it} \cdot \sum_l y_{il} \right) \right] \tag{50}$$

$$WEIGHT_s = \begin{cases} N + 1 - N_AD_s & if\ N_AD_s = N_D_s \\ 0 & otherwise \end{cases} \tag{51}$$

Finally, the scenario probability is defined in Eq. (52).

$$P_s = \frac{WEIGHT_s}{\sum_s WEIGHT_s} \tag{52}$$

### 3.3. Model complexity and benefits of the formulation

The formulation proposed in Eqs. (1)–(52) (from here on, FullP) is an endogenous stochastic nonconvex mixed-integer nonlinear model. The nonconvexities stem from the decision-dependent probability defined in Eq. (52). The nonlinearity comes from the scenario weight formulation (51), which is successively multiplied by integer variables in the operational costs' terms. The scenario probability calculation is a decision-dependent uncertainty set, which means the uncertainty depends on first-stage decisions (i.e., the primary suppliers to select and facilities to establish). With different first-stage decisions, different scenario realizations will be included in the optimization.

The introduction of this additional complexity allows modeling complete risk aversion of the decision maker. The core lies in testing categorically possible scenarios where the SC's active sites (both suppliers and facilities) can face disruption events, removing all disruption-free scenarios so as not to test business-as-usual situations. As the literature highlights, getting reliable data to describe disruption probability is often difficult while valuable historical data might not be present, leading decision makers to adopt subjective probabilities based on experts' opinion [38,43,80,88]. As a consequence, most of the studies model disruption and scenario probabilities through the random generation of a set of data [3] that does not ensure a complete risk-averse setting.

In this study, the scenario generation and reduction procedure allows overcoming the probability definition as it is calculated through the number of disruption events happening in a scenario, where a higher number of disruption events results in a lower scenario probability. Moreover, the additional complexity introduced through this specific probability formulation is threefold. First, it allows reliance on a completely unbiased method for probability formulation. Second, it avoids requiring the decision makers to define specific disruption risk probability: evaluating and using discrete probabilities for disruption events can be difficult given the lack of available data, and this can lead to results that are not representative of reality [33,78,84]. Third, it ensures complete risk aversion and considers only scenarios with disruptions at selected sites.

### 3.4. Heuristic solution procedure

Models with decision-dependent probabilities are very rare in the literature and difficult to solve [46,59]. To tackle these computational challenges, we developed a two-step solution method using elements that can be found in the concepts of Benders decomposition and local search [60,62]. FullP is divided into two problems, as follows:

- The first problem (called MasterP) corresponds to a reduced mathematical formulation with only one disruption-free scenario (with unitary probability).
- The second problem, called the sub-problem, consists of the complete formulation introduced in Eqs. (1)–(52) with fixed primary suppliers and established facilities coming from MasterP.

MasterP considers one scenario only; therefore, it is a mixed-integer linear mathematical problem. Through an optimizer (i.e., CPLEX), MasterP is solved, and the selected primary suppliers' and opened facilities' indices are passed to the sub-problem. As such, scenario probabilities in the sub-problem are fixed, and the

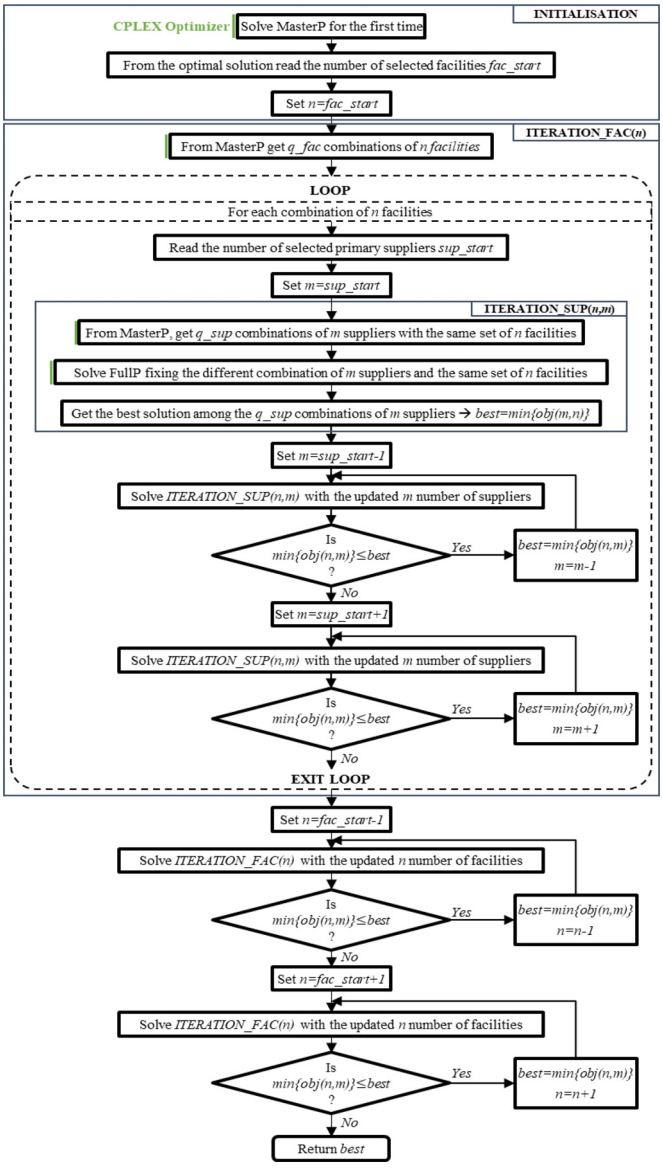



**Fig. 2.** Heuristic procedure.

nonconvexities and nonlinearities can be solved by transforming FullP into a mixed-integer linear program that could be solved by the most common optimizer. Only the locations of the suppliers and facilities are forced. The optimal capacity level is determined through the solution to the sub-problem. This master/sub-problem procedure is applied iteratively in a local search framework (Fig. 2).

The initialization consists of the first execution of the decomposition procedure. A number *n_fac* of opened facilities and a number *n_sup* of selected primary suppliers are set as starting points. By including the appropriate additional constraints to MasterP, different combinations of facilities and suppliers are obtained and successively tested in the sub-problem. The number of combinations for the same number of facilities is identified by the hyperparameter *q_fac*, while the number of combinations for the same number of suppliers is described by the hyperparameter *q_sup*.

The procedure continues by changing the number of suppliers for each combination of facilities and the number of facilities to establish. The iterations cease once the subsequent change in the number of suppliers/facilities does not improve the solution. This iterative procedure generates numerous runs of several mixed-integer linear programs. In the MasterP run, the reduced linear program benefits the removal of multiple scenarios, resulting in a single one. Successively, FullP is significantly smaller than the original formulation (1) as the scenario set is reduced based on the selected suppliers and facilities, which are forced, leading to a reduced scenario set. More details about the problem size comparison are available in Appendix 2,Tables A2 and A3.

**Table 2**
Parameters of randomly generated instances .

| Parameter | SMALL | MEDIUM | LARGE |
|---|---|---|---|
| K | $2 \le K \le 5$ | $6 \le K \le 10$ | $11 \le K \le 20$ |
| OF | $0 \le OF \le 1$ | $0 \le OF \le 2$ | $0 \le OF \le 3$ |
| NF | $2 \le NF \le 5$ | $6 \le NF \le 10$ | $11 \le NF \le 20$ |
| J | $5 \le J \le 10$ | $11 \le J \le 50$ | $51 \le J \le 100$ |
| L, P | $2 \le L \le 3$ | $4 \le L \le 7$ | $8 \le L \le 12$ |
| $D_{jt}$ | $\sim U(100,\ 1,000)$ | | |
| $CPS$ | $\sim U(40,000,\ 80,000)$ | | |
| $CPQ_k$ | $\sim U(10,\ 50)$ | | |
| $CBS_k$ | $\sim U(8,000,\ 16,000)$ | | |
| $CI_i$ | $\sim U(80,000,\ 160,000)$ | | |
| $CC_i$ | $\sim U(10,\ 50)$ | | |
| $CP_k$ | $\sim U(1,\ 10)$ | | |
| $\gamma$ | $\sim U(1.4,\ 2)$ | | |
| $CT$ | $\sim U(0.01,\ 0.03)$ | | |
| $CH$ | $\sim U(0.1,\ 1)$ | | |
| $\varphi$ | $\sim U(15,000,\ 25,000)$ | | |
| $\theta_i$ , $\theta_k$ | $\sim U(0.2,\ 1)$ | | |

## 4. Computational and comparative analysis

In this section, we present the computational analysis of the introduced nonconvex nonlinear mathematical model to discuss the proposed heuristic solution procedure and the influence of the parameters. All the algorithms were coded in the Python programming language on the Jupyter Notebook platform and run on a PC with a 3.6 GHz Intel Core i7–4790 processor and 16 GB memory with CPLEX 12.1.

### 4.1. Computational experiments (optimal vs. heuristic)

The computational experiments have been performed with a set of randomly generated data (Table 2) to test the proposed heuristic procedure and compare it with a commercial solver able to handle nonconvex mixed-integer nonlinear problems, i.e., the BARON Solver [54].

For one event per scenario ($N = 1$, Table A2), small instances can be optimally solved by the BARON Solver. The optimal solutions are obtained after 1000 sec (approx. 15 min) on average, while convergence takes averagely 2.5 h. The best heuristic solutions obtained are always equal to the optimal value from the BARON Solver, while the “CPU time to best” improves significantly at approximately 1 second per instance on average. For medium instances, BARON converged for 8 out of 10 datasets, with an average “CPU time to best” of about 2.5 h and a CPU convergence time of around 5 h. The heuristic obtained the same objective value of the optimal solver, even for non-converged instances. The performance in terms of computational time is highly improved compared to the optimal solver, with 7.5 sec to reach the best value on average. For large datasets, the BARON Solver is not able to converge; 2 out of 10 instances were not able to run because of system memory limitations. The proposed algorithm is always able to obtain a solution even in a reasonable computational time (i.e., approx. 120 sec). Among the 8 converged instances, 50% obtained the same objective value; the heuristic procedure was able to obtain a better solution for the other half of instances with a 0.5% average improvement.

Considering two events per scenario ($N = 2$, Table A3), the problem size increases substantially, and the BARON Solver was

only able to converge for small instances with an average time of approximately 65 min (for the optimal solution) and a convergence time of about 3.5 h. On the other hand, our heuristic algorithm obtains the same optimal solution with a "CPU time to best" of 1.5 sec on average. For medium- and large-sized datasets, our hardware's performance began to suffer with the more complex problems: 60% of the medium-sized instances and all the large-sized instances ran out of memory, and the solver was no longer able to run. On the contrary, the developed heuristic procedure is always able to obtain a solution in a reasonable computational time. For medium-sized datasets, our algorithm takes an average of 135 sec to obtain the best solution. Furthermore, for large-sized instances, the best time equals 40 min on average.

The "CPU time to best" solution directly depends on two main factors: $N$ and the number of iterations. Especially for large-sized instances, the number of iterations plays a crucial role in finding a good solution in a reasonable amount of time. In fact, the starting point of the heuristic procedure (based on *n_sup* and *n_fac*) depends directly on the dataset and not on the modeler. Therefore, if the starting point is inadequate, more iterations are needed, and a longer computational time is required. Furthermore, the greater the $N$, the greater the computational time. For the majority of the tested datasets, the number of iterations for the best solution is the same for $N = 1$ and $N = 2$; the increased complexity (and time for the best solution), therefore, mostly comes from the problem size, which increases with a greater number of scenarios.

In summary, the solutions of the nonlinear model through the heuristics are reached within reasonable amount of time despite the necessity for many runs of different linear mixed integer programs. For small and medium sized instances, the heuristic best solution coincides with the optimal solution obtained through convergence of BARON Solver. The heuristic algorithm always outperforms the nonlinear solver in terms of the time to reach the best solution and the ability to solve an instance. Furthermore, it is able to acquire improved objective function values for medium-/large-sized instances. Therefore, we consider the proposed solution procedure validated and appropriate for the analyzed problem.

### 4.2. Comparative analysis

In this section, the effectiveness of the specific components of the resilience portfolio included in the formulation—namely, multi-sourcing, backup suppliers, protection systems, and recovery actions—are investigated through an extensive statistical analysis. To study the main effects of the proposed resilience strategies, a $2^k$ fractional design method was employed. First, the objective value (i.e., total costs) for all possible combinations of resilience strategy effects was measured. Second, the obtained results were analyzed through several factorial plots in which we can see the main effect of considering ("$+1$") and ignoring ("$-1$") each strategy on the marginal response (i.e., total costs) average. Fig. 3 presents the results for different configurations of disruption occurrence.

In Fig. 3, the insignificant effect is represented by a straight line; otherwise, the positive main effects mean that the consideration of the strategy positively affects the response—i.e., it reduces the value of the total cost. Considering the suppliers' disruption, the resilience strategies involved are *multi-sourcing* and *backup suppliers* (protection systems and recovery actions do not influence primary suppliers). The *multi-sourcing* strategy achieved a positive main effect even for small-/medium-scale disruption.

The *backup supplier's* strategy registered a positive effect for disruption with high magnitude on suppliers' operability. *Recovery actions* registered positive and increasing main effects for small- and medium-scale disruption, after which it slightly reduced its effectiveness for high-scale disruption. For disruption with high impact on facility operations, the *protection systems* resilience strategy is the most effective. The *multi-sourcing* strategy maintained a moderately positive effect even for facilities' disruption. In fact, multi-sourcing allows to better follow changes in procurement. Generally, small- and medium-scale disruption can generate small variance and smaller main effects rather than high-scale disruption. In addition, the biggest effect was assumed via a multi-sourcing strategy in high-scale disruption at suppliers, followed by *protection systems* in high-scale disruption at facilities.

## 5. Real-case application

Our model was applied to a real-case application of an international SC in the plastic components industry. The network is composed of 50 customer zones, 19 possible locations for facilities, and 11 possible suppliers. The considered time horizon was 1.5 years and was modelled through 36 discrete periods (i.e., one period was equal to two weeks), in which disruptions were modelled starting at the second period ($t = 1$). Both the primary suppliers' contracts and the facilities' capacities were modelled through 14 capacity levels (the highest level allowed one supplier/facility to cover all the demand). Furthermore, five different levels of protection investment were included with constant ranges from 20% to 100% of the investment cost, where the highest protection level was presumed to reduce disruption impacts by 80%. The holding costs were 0.52€ per period per unit of product, and the unitary transportation costs were equal to 0.022€ per unit of distance (km). Finally, the penalty ratio for purchasing from backup suppliers was 1.6. As shown in other studies [49], the penalty cost applied for the missing units of demand was set to $1E + 5$ to model the situation in which losing the customer's demand is extremely costly. Given this assumption, the main objective of the analysis was to determine the most efficient network design and associated supply–demand allocations subject to a service level of 100% as a resilience indicator under several disruption scenarios.

As the disruption occurrence and magnitude are not easy to estimate precisely and they are intrinsically uncertain, we attempted to understand how fluctuations in this parameter impact the optimal locations and network structure of the SC. The model results were first analyzed in terms of cost and network topology for different configurations of disruption occurrence, magnitude, and duration. Subsequently, we demonstrated the efficacy of the resilience investment included with a reduced formulation removing the possibility of investing in backup suppliers, protection systems, and recovery actions (called *no-resilience action*). The detailed results are presented in Table A4 in Appendix 3.

### 5.1. Nominal design

As a starting point, the mathematical model was applied without disruption consideration ($\vartheta_k = \vartheta_i = 0$) in a *nominal scenario*. The optimal design is depicted in Fig. 4. Two primary suppliers and two facilities were established with a contracted/established capacity equal to 4900.

### 5.2. Analysis of the impact of suppliers' disruption

Next, we analyzed supplier disruption impacts and SC reconfiguration reactions. Focusing on the *complete formulation* results, suppliers' disruption can have a greater impact on SC costs than facilities' disruption. As observed in Appendix 3, suppliers' disruption could increase the total cost of the optimal design from 2% with small-scale interruption up to 17% considering full supplier failure. In contrast, facilities' disruptions registered an increase in total cost from 2% with small-scale short-duration disruptions to 10% considering high-magnitude long-term disruptions. The disruption effects propagate downstream, with ripple effects occur-

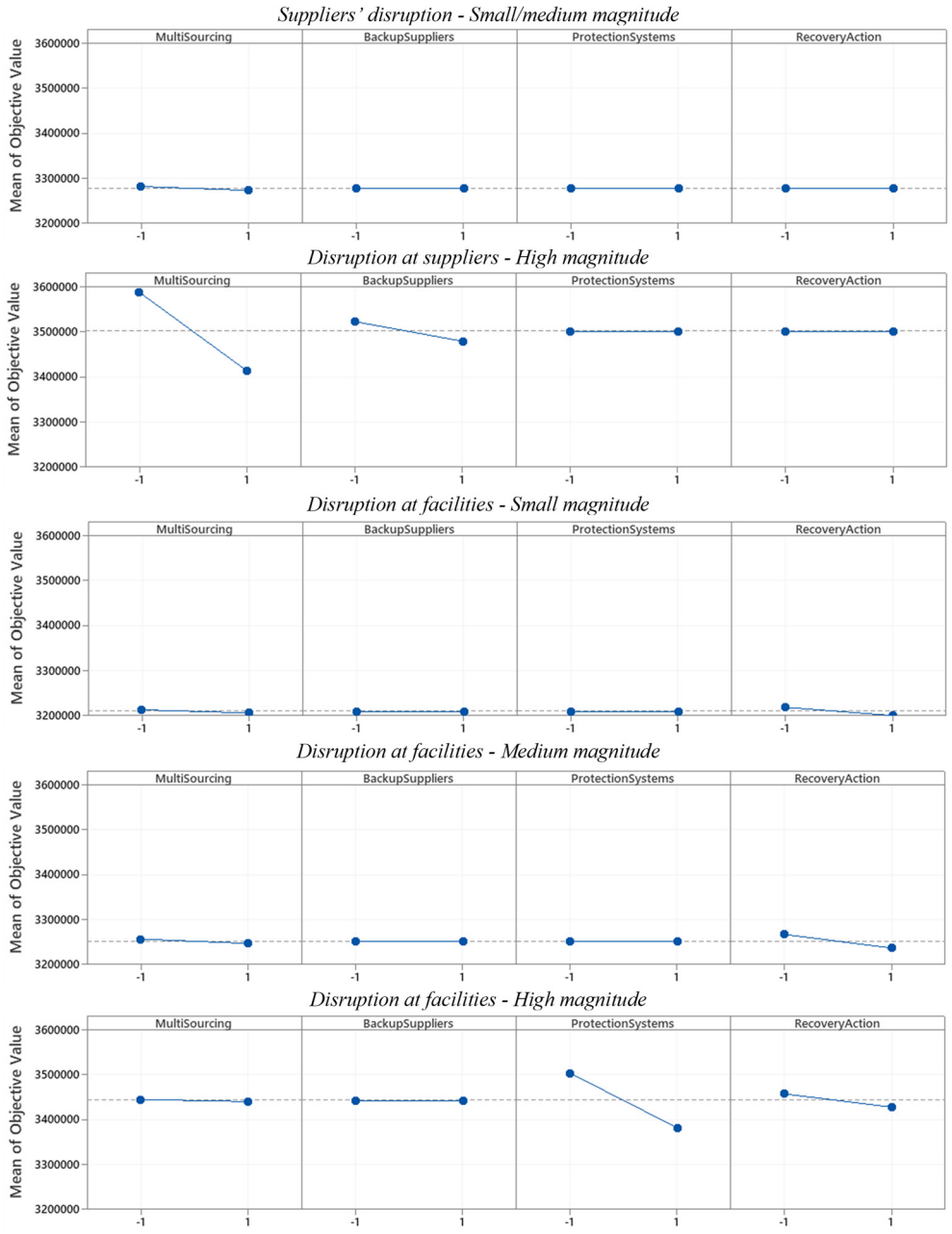



**Fig. 3.** Factorial plots for resilience investment analysis.

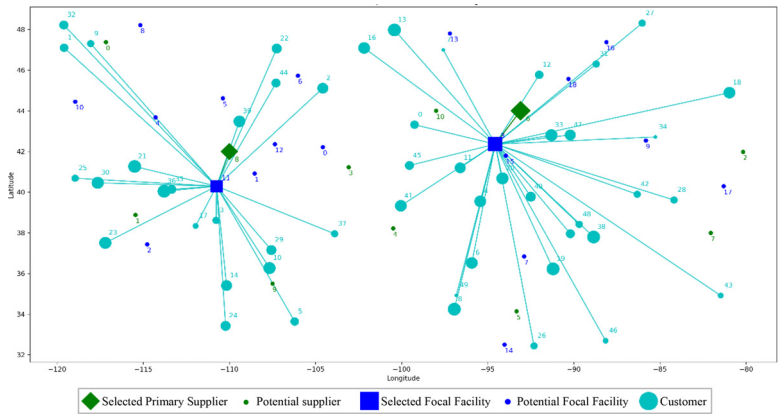



**Fig. 4.** Nominal SC design (markers' size is proportionate to material flow).

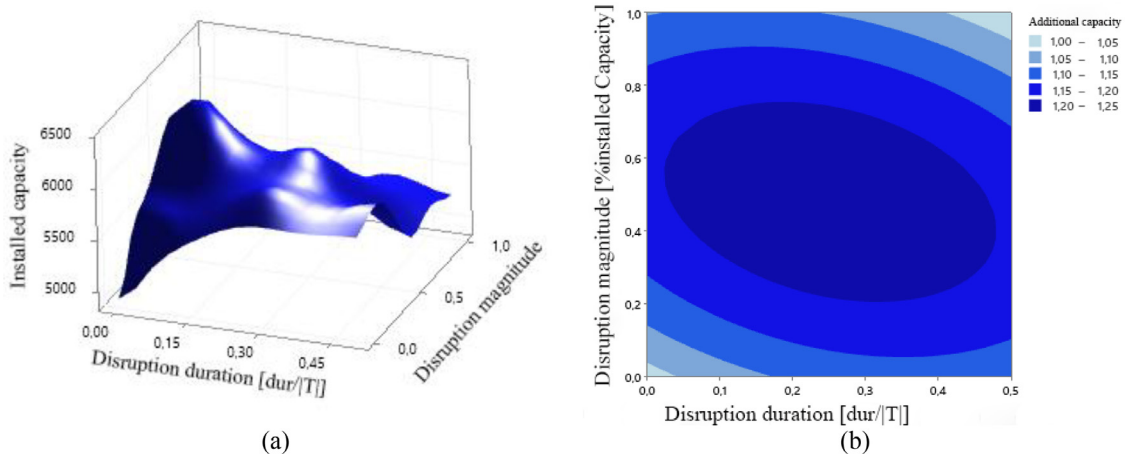



**Fig. 5.** Facilities' capacity evolutions based on experimental results (a) and polynomial regression (b).

ring throughout the SC and leading to higher impacts on total costs. To hedge against suppliers' disruption, the SC needs to increase the contracted capacity in respect to the situation with no disruption consideration. Considering a small value of risk aversion ($N = 1$), the contracted capacity increased from 4900 to 5950 and 6300 for small-, medium-, and high-severity interruptions, respectively. More importantly, the network topology evolved in a more fragmented and decentralized structure by maintaining quite the same level of contracted capacity and increasing the number of suppliers from two suppliers with a disruption impact of 20% of capacity (average contracted capacity per supplier = 2950) up to two suppliers considering a disruption impact of 100% of capacity (average contracted capacity per supplier = 1260). This evolution aimed to increase the robustness of the upstream network through larger contracted capacities and enhance flexibility through supplier fragmentation. Backup suppliers' agreements were employed only with high-scale disruption. By increasing the risk aversion of the decision maker, an additional increase in the contracted capacity was observed, followed by an increase in the number of suppliers (i.e., fragmentation).

### 5.3. Analysis of the impact of disruptions at facilities

With regard to disruptions at facilities, disruption consideration immediately leads to an increase in the established capacity from 4900 to 5600. With low-magnitude disruption events, increasing the capacity is the first mitigation plan that allows retaining sufficient available production/distribution capacity and maintaining the material flows without significant changes in SC configuration and topology. Recovery actions can be efficiently deployed for short- and medium-term disruptions without redundancy investments. In contrast, for long-term disruption, proactive mitigation plans are necessary to completely fulfill the customers' demands. For a medium-magnitude disruption effect, the SC evolves into a more decentralized structure. For short-term disruption, increasing the number of facilities and capacity together with recovery actions after disruption occurrence renders the most efficient SC configuration. With an increase in the disruption duration and risk aversion attitude, the protection systems' investment becomes an efficient mitigation plan to increase SC resilience. This result is successively confirmed with high-impact disruption events, where the optimal SC design is marked by a lower level of capacity with respect to the network obtained with low-impact disruption events.

We illustrate these findings in Fig. 5. Figure 5(a) plots the established capacity obtained in relation to the disruption magnitude and duration from the experimental results presented in Table A4. As observed in the surface plot [Figure 5(a)], the capacity peak does not correspond to the highest disruption impact or duration, but it is associated with the medium values of the two parameters. Figure 5(b) presents a contour plot of the established capacity as a ratio between the nominal capacities obtained by applying a polynomial regression of the experimental results. As shown in Figure 5(b), the highest capacity increase can be observed in the middle of the plot, with medium disruption magnitude and duration, while the extreme points on the graph present a lower percentage of capacity increase.

The two plots in Fig. 5 show that an excess of capacity (SC robustness) does not always lead to the most efficient solution to disruption risk. Finally, changes in the facilities' locations and numbers shift the supply-side network to a more fragmented supplier base. However, in contrast to disruption from only the supply side, the contracted capacity from primary suppliers did not face any significant changes, and it remained the same as in the nominal scenarios.

### 5.4. Analysis of the impact of simultaneous suppliers' and facilities' disruption

Considering disruptions simultaneously at both suppliers and facilities, the SC required a high level of robustness and resilience. Focusing on the numbers and locations of suppliers and facilities, with these disruption configurations, the SC structure was similar to that obtained by considering suppliers' disruption. This result further confirms that supply-side disruptions have a greater impact on SC network topology than facilities' disruption. Moreover, focusing on the best mitigation and contingency plans, the main insights obtained by the two analyses were generally confirmed. For low-magnitude disruption, increasing robustness through extra capacity and restoring lost capacity through recovery actions is the most efficient resiliency plan to maintain a 100% service level. Furthermore, for high-magnitude disruption events, protection systems maintain a fundamental role in efficiently managing disruption at facilities. In addition, the more important role of backup suppliers was noted. The optimal SC network, obtained with a disruption impact $\theta_k = \theta_i = 1$, was described by four primary suppliers, two backup supplier, and four facilities. This configuration maintained a more balanced/less skewed flow from primary suppliers, fulfilling local and temporary peaks through backup suppliers.

### 5.5. Robustness vs. resilience

In this section, we first remove the recovery components of the resilience portfolio—i.e., backup suppliers, protection systems, and recovery actions (i.e., *no-resilience action*), reducing the analysis to robustness only—and then compare the results with the *complete formulation* (i.e., both robustness and recovery). *No-resilience action* entails increasing the capacity and number of suppliers/facilities to hedge against disruption risk. The mathematical formulation of *no-resilience action* consists of the same problem presented in Section 3, where the second term (backup supplier), the fifth term (protection/flexibility systems), and the thirteenth term (recovery actions) are removed from the objective function, with their related dependent terms in constraints. Table A4 in the appendix presents the design obtained by the two formulations. The *complete formulation* leads to significant savings in terms of total cost. The specific resilience features may prevent overcapacity and/or an increase in the number of facilities, which may result in a total cost reduction of up to 12%. In general, the main differences lie in the greater investment cost of the robustness-oriented formulation.

An additional test was performed by selecting the two main designs obtained with two sets of disruption configurations derived from the two formulations. The SC configurations from the *complete formulation* were called the *resilient design*. The results from *no-resilience action* were called the *robust design*. The *nominal design* corresponded to the SC resulting from the optimization with no disruption consideration. The details in terms of the primary/backup suppliers and facilities of the selected design are provided in Table A5 in Appendix 4. The results are summarized in Table 4.

Table 4 shows that the nominal design only performed well in the nominal scenario; even with the minimum impact of a disruption event, the SC was not able to fulfill all customer demands. On average, the demand loss was 5%, 25%, and 50% for small-, medium-, and high-impact disruption events, respectively. In the nominal scenario without disruptions, the increases in total costs from the resilient design were 3% for Design #1 and 11% for Design #2. The increase in investment costs caused by higher resilience led to a more efficient network, even in nominal scenarios allowing operational cost savings of 6.2% and 11% in Design #1 and Design #2, respectively. Design #1 performed efficiently and could sustain low-magnitude short-term disruption. However, when the disruption impact increased, the SC was unable to meet the customers' demands. Design #2 was always ready to hedge disruption impacts. A comparison between resilient and robust designs shows that the resilient designs always outperformed the robust ones, except when considering only medium- and high-impact suppliers' disruption. In this context, the backup supplier agreement established in Design #2 allowed the material flows to be maintained at a higher cost and caused an increase in operational costs of around 8%–10%. In contrast, the backup suppliers performed well in the facilities' disruption case and in the suppliers' and facilities' simultaneous disruption case by ensuring total cost savings of up to 11%.

Fig. 6 presents the comparison between the SC structures obtained by the *complete formulation* [Figure 6(a)] and *no-resilience action* [Figure 6(b)] with a high-impact ($\theta_k = \theta_i = 1$) long-term ($dur = 18 = T/2$) disruption occurrence at both suppliers and facilities. The maps show the established primary/backup suppliers and facilities as well as the material flows registered in all considered scenarios. As shown in Fig. 6, the robust design led to highly complex material flows. The different capacity shortages that suppliers/facilities face as a result of disruption occurrence required the SC to adapt and reassign the "normal" and uniform material flow based on the remaining available capacity, generating an unbalanced upstream flow that propagates in the cascading downstream. The investment in protection systems enables the basic concepts and policies of material distribution to be maintained, reducing reassignment activities at the customer flows. In addition, the backup suppliers mitigated the supply shortages and the skewed material requests, assuring a less entropic system. In Resilient Design #2, no clustering was identified in terms of which facility serves a group of customers. Instead, the robust design presented highly congested flow lines with numerous intersections and efforts to completely fulfill customer demand. Even if the total cost savings from resilient to robust design are minor (3%–4% of the total cost by considering suppliers' and facilities' disruption simultaneously), the increase in management complexity could be extremely high, favoring the resilient solution.

## 6. Managerial insights

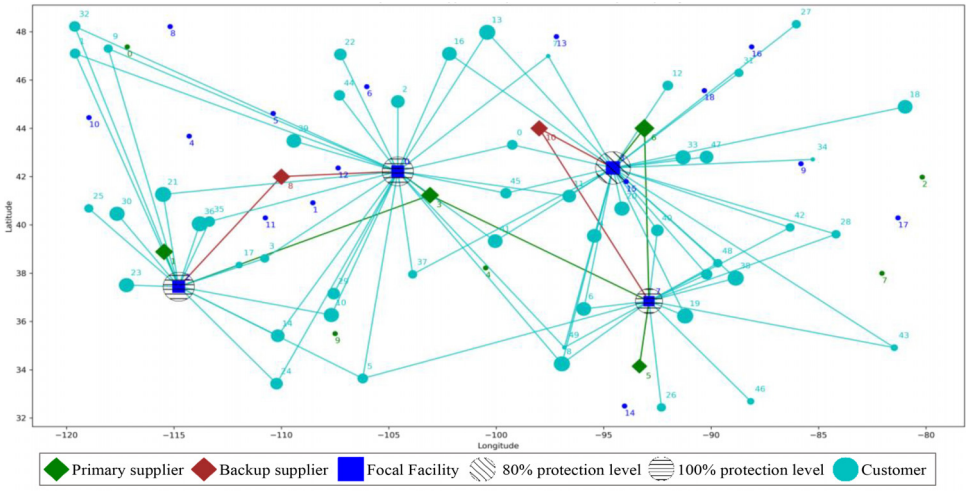
In this study, we addressed the practical challenge of combining the available resilience assets—such as backup supply, facility fortification, capacity flexibility, multi-sourcing, and rerouting in an efficient way—into a resilient SC portfolio that incurs the lowest costs at the preparedness stage and allows for effective recovery if a real disruption happens. Our results show that investments into SC preparedness and the resulting efficient resilience portfolio can even increase SC efficiency in business-as-usual (i.e., disruption-free) scenarios, contributing to understanding and implementing lean resilience.

In Table 5, we summarize our data-driven key findings and generalize associated managerial insights deduced from the results of computational and comparative analysis and case-study application as well as discuss them in depth further into this section. We now detail the analysis at the network, supplier, and factory levels as well as extrapolate our major findings and generalized managerial implications to the COVID-19 pandemic Table 3.

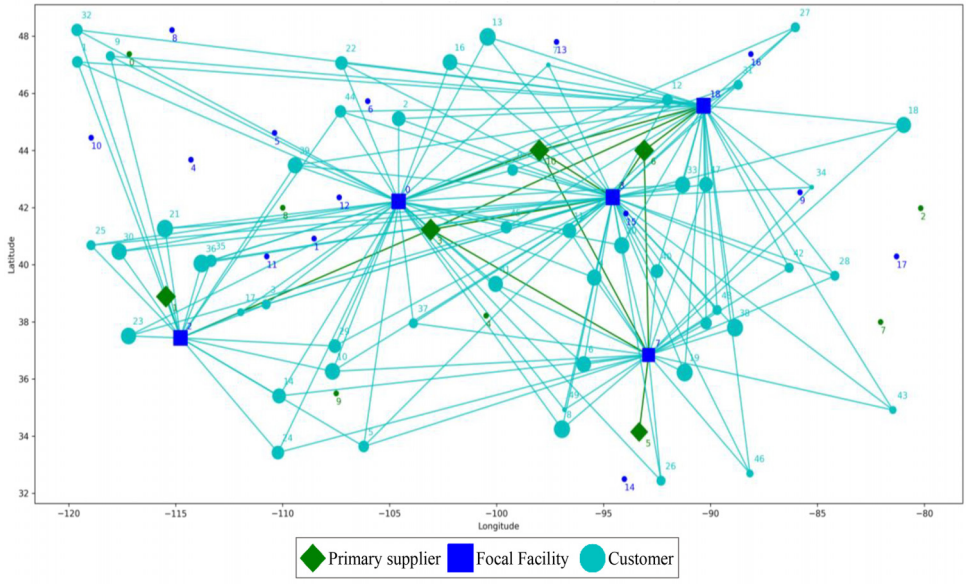
### 6.1. Efficient resilience for network design

A traditional way of designing resilient SCs by extending efficient network design with some redundancies (e.g., risk mitigation inventory and capacity buffers) frequently results in high costs and resource consumption. This way of building resilience is expensive since many of these redundancies will just be waiting for use in an emergency case without creating any value at business-as-usual times. In this setting, SC managers may be reluctant to undertake large increases in SC costs unless they become aware of substantial improvements in both resilience and efficiency that can be attained with minimal increases in SC costs [87,90,91].

Realistically, company managers may not accept an SC with high resilience investment costs just to hedge against very rare disruption risks. However, our study shows that a good trade-off is achievable with only minimal investments. Agile, flexible, and reconfigurable SCs can be efficient, and they are also resilient through structural and process variety [76]. In this setting, resilience is not built upon efficiency; resilience is embedded into every business-as-usual operation and a part of efficiency (i.e., lean resilience). Examples include omnichannel distribution systems, multiple sourcing, diversified logistics networks, and flexible production lines, which are good for both efficiency and resilience [35]. For example, Design #1, with a low level of redundancy, obtained an increase of 2.8% of total costs in the nominal scenario; however, it could efficiently sustain small-/medium-magnitude disruptions. Given a medium level of redundancies and a high level of flexibility/protection systems, Design #2 had an 11% increase in total cost compared to nominal scenarios while it was able to fulfill customer demand in each disruption scenario. Further, resilient designs were more efficient in terms of operational costs in nominal scenarios when no disruption occurred. As such, our optimization




(a) Resilient design.




(b) Robust design.

**Fig. 6.** Resilient Design #2 (a) and Robust Design #2 (b) with a high-impact long-term disruption occurrence at both suppliers and facilities (the size of the points is proportional to the installed capacity).

approach can help allocate resilience investments so that they can be used in everyday business operations to generate value. Therefore, SC managers could be more incentivized to include disruption risk and resilience features in SC design decisions as this could increase both competitiveness in business-as-usual scenarios and resilience. This result contributes to the literature on transforming resilience from an expensive spend to a value-creation asset [16,32].

### 6.2. Suppliers' disruption: main effects and resilience plans

Supply disruptions were confirmed to be more critical and impactful in terms of SC cost performance and network structure (Qi et al., 2010; [40]). In this setting, multi-sourcing is essential to dynamically adapt supplies among the remaining available capacity while adding value-creating flexibility at business-as-usual times. Our results show that the engagement of more raw material suppliers in a resilient SC allows the unaffected suppliers to compensate for the supply shortage in disruptions. Additionally, it reduces the dependency on a single source that could be highly efficient but more vulnerable to disruption . Furthermore, the model found that using backup suppliers' agreement was more justified as the disruption events became more impactful (high-magnitude events of long-term duration). Backup suppliers are useful to increase flexibility on the supply side of an SC, to cover completely disrupted primary suppliers, and to guarantee a more balanced material flow.

In the COVID-19 pandemic setting, this finding means that backup suppliers, especially outside the main SC footprint, could assure supply continuity in the long term and tackle possible closures caused by local government restrictions. Moreover, the optimal design results found that suppliers' disruptions had a strong effect on facilities' locations and capacities. The real-case application recognized that the increase in primary suppliers' dispersion led to an increase in the number of facilities. Here, the material flow shifts among the available suppliers, causing an un-

**Table 3**
Key findings and managerial insights.

| Results of computational and comparative analysis and case-study application | Generalized managerial insights |
|---|---|
| • The nominal design is not able to completely fulfill the customer demand even under small-scale, short-term disruptions, and it could generate up to 50% of loss in demand in the most pessimistic scenarios (high-magnitude disruptions at the supply side). | • The consideration of disruptions in the design phase is essential to avoid demand losses and render immense cost reductions. |
| • Small increases in the total cost from resilience plans allows hedging against small/medium disruptions; resilient designs are even more efficient in terms of operational costs in the nominal scenario when no disruption occurs. | • Resilience does not need to appear as an expensive paradigm. A good tradeoff between resilience and efficiency is achievable with an optimization of the resilience portfolio using our approach. |
| • Supply disruptions are more critical and impactful in terms of SC cost performance and network structure. | |
| • An increase in primary suppliers' dispersion leads to an increase in the number and capacity of facilities. | • SC resilience managers should especially focus on the upstream network as suppliers are, in general, external stakeholders with their own decision system. |
| • Installed capacity at facilities should follow the contracted capacity with primary suppliers. | |
| • The differentiation of procurement strategy through a larger supply base allows to compensate efficiently for supply shortage (adapting the supply-material flows). | • Adopting dual/multi-sourcing in the upstream network is recommended. |
| • Backup suppliers are useful to increase flexibility on the supply side of an SC, to cover completely disrupted primary suppliers, and to guarantee a more balanced material flow. | • Activating backup suppliers is reasonable especially for high-magnitude disruptions. |
| • Recovery actions are highly effective for short-term, small-scale disruption. | • A focus on being rapid in reallocating the material flow and successively restoring the lost capacity for covering small-scale disruption is recommended. |
| • The no-resilience action design requires a capacity increase of 50% more than the established capacity in complete formulation designs. | • High redundancy levels are not recommended as they are too expensive and do not bring the required flexibility to hedge high-scale disruptions. |
| • As disruption events become more severe and last longer, investment in protection/flexibility systems becomes essential to efficiently increase SC resilience without requiring a high increase in the established capacity. | • For hedging against severe disruption events, SCs need to improve their asset focusing on increased flexibility and enhanced differentiation with the goal to design available alternatives in their operations instead of boosting robustness. |

balanced upstream flow that propagates, cascading to the downstream flow and increasing facilities' installed capacity. Therefore, suppliers' disruption needs higher attention as it affects the SC and could lead to an immense increase in total costs and lost demand [23]. In general, suppliers are external stakeholders with their own decision systems. For this reason, SC managers might consider alternative management strategies, such as integration, collaborations, and differentiation. In general, supply disruption mitigation principles should leverage two sides: preparedness among supplier bases and adaptability at facilities. This could enhance the operations' fulfillment in suppliers' disruption scenarios and mitigate the ripple effect.

### 6.3. Disruptions at facilities: main effects and resilience plans

Based on the results of the facilities' disruption modeling, robustness and resilience principles need to be included simultaneously in the SC design. The robustness principles mainly look at planning redundancies to reserve excess capacity and be able to meet the customers' demand in any situation. However, the redundant capacity for hedging against high-impact disruption events could be extremely high. In the real-case application, the *no-resilience action* design required a capacity increase of 50% more than the established capacity in the *complete formulation* designs. In addition, opening a new facility/distribution center implies that another point in the SC may be subject to disruption and therefore lose its operability. Finally, the presence of more sites in the SC could increase complexity in terms of materials management and logistics procedures. Therefore, resilience and flexibility principles play a crucial role in efficiently hedging against disruption risk.

The results show that recovery actions are highly effective for short-term small-scale disruption. During disrupted periods, the SC adapts the material flows based on the remaining available capacity. Subsequently, when disruptions end, recovery actions are performed to restore the required capacity and return to a more efficient distribution flow. As the disruption events became more severe and lasted longer, protection/flexibility systems became essential to efficiently increase SC resilience without requiring a high increase in the established capacity. In the sensitivity analysis experiments, the highest disruption magnitude optimal design reduced the installed capacity and prioritized investing in protection systems and flexibility compared to less impactful disruptions.

These findings are transferable to a disruption scenario similar to the COVID-19 pandemic, where companies can be subjected to a long and unexpected halt in their operations. Flexibility investment may be seen as a winning strategy to reduce the economic impact of the disruption and to maintain operations and sustain business or enter other markets based on need. In this context, flexibility can be achieved by using more adaptable manufacturing systems or risk-pooling strategies, postponement, and component commonalities, which echoes and quantitatively confirms the conceptual results offered by Sodhi et al. [81]. The excess capacity is smaller and more easily accepted by upper-level management. In addition, unused capacity could be smartly employed at business-as-usual times for other purposes (e.g., prototyping, new product development, machine rotation) to reduce failures, maintenance, and operational risks.

## 7. Conclusions

In this study, we engaged in answering a relevant and novel research question with a strong practical underpinning: how to add resilience components into SC network design so that it incurs the minimum costs at the preparedness stage and allows for efficient and effective recovery in case of a real disruption? We referred to the combination of proactive and reactive actions such as backup suppliers, extra capacity, facility fortification systems, multi-sourcing, and rerouting material flows as the *resilience portfolio* and proposed a non-trivial model and an original heuristic algorithm to optimize resilience portfolio efficiency with the consideration of preparedness and recovery costs.

Technically, our paper offers a new endogenous stochastic non-convex mixed-integer nonlinear mathematical model for the de-

sign and planning of a resilient SC in a three-echelon, multi-period, single product setting. Our formulation proposed a decision-dependent scenario probability formulation that considers a full spectrum of disruption scenarios with the influence of actual activated sites (suppliers and facilities) while capturing the risk aversion of the decision maker. Disruption events were included, and they could partially or completely reduce the available capacity at both primary suppliers and facilities. The model objective was to increase resilience at the minimum costs by determining the optimal primary suppliers' and facilities' locations, capacity and number, supply- and demand-side (re)allocation of material flows, and levels of protection capacity. Distinctively, proactive and reactive resilience actions were integrated to hedge against disruption risk. Our findings identified important relations between disruption duration/magnitude and the efficiency of preparedness and adaptation strategies. The outcome of this research can be instructive for SC managers when deciding on the investment in redundancy allocation and the efficient use of this redundancy during recovery.

Limitations exist, as with any study. We considered a single-product setting while multiple products could increase the comprehensiveness of the analysis. We assumed the existence of fully reliable suppliers, which is a common assumption in the literature but not always possible in real SCs. We did not consider any restrictions on the available budget for building a resilient portfolio; this can be used to examine capital-constrained settings. In our model formulation, disruption and recovery are considered as events with immediate impacts, which is also a common assumption in the literature but can be different in some settings (e.g., the COVID-19 pandemic) when capacity shutdown and ramp-up are better to model as time functions [7,37,38]. Furthermore, the proposed heuristic is tailor-made to the mathematical formulation proposed in the article (i.e., supply chain network design) and could be adapted to models with a similar complexity setting (i.e., endogenous stochasticity, decision-dependent probabilities) with some modifications. Moreover, the decision-dependent probability formulation allowed the risk-aversion attitude while the disruption probabilities are, however, equal among the considered points. Our approach is particularly suitable for decision-making situations when all sites/locations have equal disruption probabilities. This is a very common case in practice since, especially in a complex network, it is difficult to differentiate disruption probabilities because of missing historical data [45,78,79]. In a generalized setting, our model can be tailored to situations with heterogenous disruption probabilities with some minor adaptation by transforming $\text{weight}_s$ to be an input parameter defined by the decision maker according to the risk exposure of the considered locations.

Several research directions may be considered for future research. One possible extension is to include specific SC components, such as manufacturing features or different product types, to provide insights regarding production facilities and a more in-depth analysis of material flows. For a better generalization of results, the model should be adapted and tested for different industrial sectors—such as food/perishable, pharmaceutical, big size, and luxury goods SCs—to derive insights from different SC configurations. Moreover, we did not consider any source of operational risk. Therefore, future research could include variations in demand or cost parameters to capture the dynamics of the markets [12]. Furthermore, the inclusion of reverse flows and closed-loop SCs could allow the analysis of the influence/mitigation potential of circular flows for product recycling and/or disposal [55,83]. The scenario generation procedure, in particular, generation of disruption probabilities needs to be justified by industrial practices. For example, in the COVID-19 pandemic setting, the regional disruption scenarios can be introduced, and the interregional disruption propagation explicitly modeled [7,74]. Next, since our approach resulted in nonlinearity of the proposed model caused by the imposed reduction of disruption scenarios, other alternative approaches such as CVaR-based risk averse optimization can be tested. When using CVaR-based optimization, business-as-usual scenarios can be neglected (if necessary) by selecting a high confidence level, e.g., 0.95, 0.99, and what is most important, linearity of the model could be preserved. As such, in order to further evaluate performance of the proposed nonlinear model, our results could be computationally compared with an equivalent linear one without scenario reduction and with the CVaR objective function. We stress that the novelty of our study is related to managerial insights rather than to computational efficiency the improvement of which can be further examined in future research. In addition, other heuristics such as learning-based algorithms can be tested [44,63]. Finally, a new context of shortage economy [36] and viability [37,39] for SC management raised by years of disruptions and crises caused by the pandemic, product shortages, and geopolitical tensions as well as positive changes driven by Industry 5.0 [10,29] can be considered in the framework of our study.

## CRediT authorship contribution statement

**Riccardo Aldrighetti:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Formal analysis. **Daria Battini:** Conceptualization, Writing – original draft, Writing – review & editing, Resources, Supervision. **Dmitry Ivanov:** Conceptualization, Writing – original draft, Writing – review & editing, Supervision.

## Data availability

The data that has been used is confidential.

## Appendix 1. Literature review

**Table A1**
Literature comparison with other remarkable works in the field.

| Authors | Resilience Modelling | | | Risk-aversion | Uncertainty Characteristics | | | Model Characteristics | | Disruption Characteristic | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Proactive Strategy | Reactive Strategy | Expected Disruption Cost | | modeling method | Uncertainty Type | Multi-Periodic | Network Echelons | Decision Location Layer | Disruption Layer | Magnitude of Failure |
| [91] | – | OR | O | – | SSO | EXO | – | 3 | 1 | 1 | Full |
| [9] | – | OR | – | RO | RO | EXO | – | 3 | 2 | 1 | Full |
| [92] | MS | OR | – | RO | RO | EXO | √ | 3 | 1 | 1 | Full |
| [25] | – | OR | – | PR | RO | EXO | – | 3 | 2 | 2 | Full |
| [84] | – | OR | – | PR | FPP | EXO | – | 3 | 2 | 2 | Partial |
| [19] | FF | CR, TRS | R, TS | SDT | SSO | EXO | √ | 3 | 2 | 1 | Partial |
| [40] | EC, MS | OR, BS | P | – | SSO | EXO | – | 3 | 1 | 2 | Partial |
| [41] | MS | TRS | TS | RO | RO | EXO | √ | 3 | 1 | 2 | Partial |
| [89] | EC | OR, CR | – | RO | RO | END, EXO | √ | 3 | 1 | 1 | Partial |
| [93] | – | OR | TP | CVaR | SSO | EXO | – | 3 | 2 | 1 | Partial |
| [66] | BS, EC, MS | OR, TRS | O, P, TS, | – | SSO | EXO | – | 4 | 2 | 3 | Partial |
| [94] | FF, MS | OR | – | – | SSO | EXO | √ | 3 | 2 | 1 | Full |
| [87] | – | OR | – | – | UT | EXO | – | 3 | 1 | 1 | Full |
| [1] | FF, MS | BA, TRS | O, TS | – | FPP | EXO | √ | 4 | 2 | 2 | Full |
| [22] | BS, EC, FF, MS | OR | O, P | – | SSO | EXO | √ | 3 | 2 | 1 | Partial |
| [95] | – | OR | O | – | SSO | EXO | – | 4 | 2 | 3 | Full |
| [52] | BS | OR, CR | O, P | RO | FPP | EXO | – | 3 | 2 | 2 | Partial |
| **This paper** | BS, EC, FF, MS | OR, CR | D, O, P, R | DDP | SSO | END | √ | 3 | 2 | 2 | Partial |

BA=Backup Assignment; BS=Backup Supplier; CR=Capacity Recovery; CVaR=Conditional Value-at-Risk; *D*=Damages Costs; DDP=Decision-dependent Disruption Probability; EC=Extra Capacity; END=Endogenous Uncertainty; EXO=Exogenous Uncertainty; FPP=Fuzzy Possibilistic Programming; FF=Facility Fortification; *O*=Opportunity Cost; OR=Operational Reassignment; *P*=Procurement Penalty; PI=Pre-positioned Inventory; PP=Postponement; PR=*p*-robustness; *R*=Recovery Cost; RO=Robust Optimization; SDT=Semi-deviation from target; SSO=Scenario-based Stochastic Optimization; TP=Transportation Penalty; TRS=Transshipment Strategy; TS=Transshipment Costs; UT=Uncertainty Theory.

## Appendix 2. Computational Experiments

**Table A2**
Computational experiments result with 1 maximum number of disruption event per scenario ($N = 1$).

| dataset | \|K\| | \|OF\| | \|NF\| | \|I\| | \|J\| | \|L\| | \|P\| | Binary Variables | Integer Variables | Constraints | BARON Solver | | | Proposed heuristics | | | | | | | Δ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Best Objective Value | CPU time (best) | CPU time (convergence) | Binary Variables (first iter) | Integer Variables (first iter) | Constraints (first iter) | Best Objective Value | CPU time (best) | Iteration # (best) | Local Search Total Time | |
| small0 | 3 | 0 | 2 | 2 | 9 | 2 | 3 | 85 | 585 | 314 | 4.29E+06 | 1.6 | 4136.6 | 70 | 312 | 240 | 4.29E+06 | 0.2 | 17 | 0.3 | – |
| small1 | 4 | 0 | 2 | 2 | 6 | 3 | 2 | 142 | 612 | 342 | 5.30E+06 | 65.2 | 8794.2 | 201 | 648 | 399 | 5.30E+06 | 0.3 | 5 | 1.4 | – |
| small2 | 2 | 0 | 3 | 3 | 10 | 3 | 2 | 170 | 780 | 379 | 4.13E+06 | 173.0 | 3967.3 | 98 | 336 | 254 | 4.13E+06 | 0.2 | 4 | 0.8 | – |
| small3 | 3 | 1 | 2 | 3 | 7 | 2 | 2 | 135 | 828 | 415 | 5.18E+06 | 50.6 | 7209.0 | 114 | 495 | 332 | 5.18E+06 | 0.1 | 5 | 0.7 | – |
| small4 | 3 | 0 | 3 | 3 | 9 | 2 | 2 | 135 | 972 | 450 | 5.36E+06 | 599.1 | 8818.7 | 66 | 312 | 240 | 5.36E+06 | 1.0 | 25 | 1.1 | – |
| small5 | 4 | 1 | 2 | 3 | 8 | 3 | 3 | 241 | 1176 | 528 | 2.94E+06 | 295.5 | 13,126.2 | 210 | 756 | 436 | 2.94E+06 | 1.5 | 30 | 1.9 | – |
| small6 | 5 | 1 | 2 | 3 | 6 | 3 | 2 | 263 | 1296 | 578 | 6.18E+06 | 2728.0 | 10,003.0 | 263 | 1152 | 578 | 6.18E+06 | 1.1 | 32 | 1.8 | – |
| small7 | 5 | 0 | 3 | 3 | 8 | 2 | 3 | 183 | 1488 | 622 | 5.42E+06 | 5104.6 | 8720.7 | 70 | 288 | 228 | 5.42E+06 | 2.7 | 48 | 3.2 | – |
| small8 | 2 | 1 | 5 | 6 | 10 | 2 | 3 | 342 | 2256 | 893 | 5.64E+06 | 434.2 | 3376.9 | 70 | 336 | 253 | 5.64E+06 | 1.3 | 11 | 1.9 | – |
| small9 | 5 | 0 | 5 | 5 | 10 | 2 | 2 | 345 | 3300 | 1080 | 5.53E+06 | 1492.4 | 12,864.0 | 30 | 126 | 152 | 5.53E+06 | 0.3 | 5 | 3.6 | – |
| medium0 | 6 | 1 | 7 | 8 | 14 | 5 | 4 | 1916 | 9324 | 2253 | 2.46E+06 | 5967.5 | 14,465.8 | 604 | 2112 | 853 | 2.46E+06 | 1.1 | 7 | 11.1 | – |
| medium1 | 8 | 0 | 6 | 6 | 18 | 5 | 5 | 1488 | 9324 | 2158 | 4.74E+06 | 10,410.7 | 23,666.8 | 930 | 4200 | 1335 | 4.74E+06 | 0.6 | 5 | 12.8 | – |
| medium2 | 7 | 1 | 6 | 7 | 16 | 5 | 6 | 1757 | 9492 | 2206 | 4.40E+06 | 2995.2 | 14,450.2 | 710 | 2808 | 1038 | 4.40E+06 | 2.0 | 7 | 22.5 | – |
| medium3 | 9 | 1 | 6 | 7 | 22 | 6 | 4 | 2289 | 14,496 | 2907 | 3.46E+06 | 8147.4 | 14,465.5 | 435 | 1512 | 697 | 3.46E+06 | 2.3 | 7 | 26.7 | – |
| medium4 | 10 | 2 | 7 | 9 | 16 | 6 | 5 | 3472 | 19,380 | 3628 | 3.68E+06 | 20,553.5 | 24,687.3 | 1115 | 3900 | 1282 | 3.68E+06 | 3.1 | 7 | 35.6 | – |
| medium5 | 9 | 2 | 6 | 8 | 31 | 5 | 5 | 2334 | 21,573 | 3748 | 2.90E+06 | 3517.0 | 14,524.2 | 624 | 3744 | 1262 | 2.90E+06 | 3.2 | 7 | 49.4 | – |
| medium6 | 9 | 1 | 8 | 9 | 25 | 5 | 4 | 2709 | 22,248 | 3862 | 2.95E+06 | 4743.5 | 20,770.9 | 604 | 3168 | 1117 | 2.95E+06 | 5.5 | 8 | 50.6 | – |
| medium7 | 10 | 0 | 8 | 8 | 30 | 7 | 5 | 3440 | 23,220 | 3980 | 4.40E+06 | 8603.7 | 21,905.8 | 964 | 4320 | 1423 | 4.40E+06 | 41.1 | 58 | 127.6 | – |
| medium8 | 9 | 1 | 9 | 10 | 36 | 6 | 5 | 3843 | 32,832 | 4944 | 2.67E+06 | 18,563.9 | ***NATC*** | 1018 | 5940 | 1669 | 2.67E+06 | 7.8 | 29 | 75.3 | – |
| medium9 | 10 | 1 | 9 | 10 | 41 | 7 | 7 | 4840 | 39,060 | 5571 | 4.44E+06 | 7279.3 | ***NATC*** | 712 | 3672 | 1430 | 4.44E+06 | 6.9 | 7 | 90.2 | – |
| large0 | 11 | 1 | 13 | 14 | 61 | 12 | 10 | 14,591 | 91,725 | 9819 | *Baron runs out of memory* | | | 3462 | 15,768 | 3373 | 4.16E+06 | 34 | 7 | 340.9 | NA |
| large1 | 12 | 3 | 11 | 14 | 85 | 10 | 8 | 12,312 | 125,658 | 12,129 | 3.52E+06 | 4450.0 | ***NATC*** | 1495 | 10,260 | 2923 | 3.52E+06 | 74.2 | 7 | 685.4 | – |
| large2 | 12 | 3 | 13 | 16 | 67 | 12 | 8 | 18,012 | 127,932 | 12,263 | 2.89E+06 | 4812.5 | ***NATC*** | 2405 | 11,550 | 2843 | 2.89E+06 | 28.1 | 7 | 391.9 | – |
| large3 | 14 | 0 | 13 | 13 | 99 | 9 | 11 | 11,021 | 141,750 | 13,536 | 3.99E+06 | 7512.8 | ***NATC*** | 2652 | 23,976 | 4722 | 3.99E+06 | 31.5 | 7 | 591.1 | – |
| large4 | 11 | 2 | 17 | 19 | 64 | 8 | 12 | 15,755 | 152,820 | 13,804 | 2.82E+06 | 12,974.5 | ***NATC*** | 1894 | 12,540 | 3034 | 2.78E+06 | 17.6 | 8 | 170.5 | **−1.4%** |
| large5 | 17 | 1 | 15 | 16 | 59 | 11 | 8 | 19,212 | 153,153 | 14,103 | 3.90E+06 | 2234.2 | ***NATC*** | 4011 | 21,462 | 4054 | 3.90E+06 | 245.5 | 64 | 645.1 | **−0.1%** |
| large6 | 14 | 1 | 14 | 15 | 96 | 10 | 9 | 14,704 | 170,172 | 14,999 | 2.59E+06 | 23,623.8 | ***NATC*** | 2826 | 23,328 | 4621 | 2.59E+06 | 6.3 | 7 | 154.6 | – |
| large7 | 20 | 0 | 18 | 18 | 59 | 10 | 10 | 22,720 | 209,874 | 17,470 | *Baron runs out of memory* | | | 4808 | 28,800 | 4864 | 1.89E+06 | 0.3 | 1 | 42 | NA |
| large8 | 19 | 1 | 19 | 20 | 63 | 10 | 8 | 25,409 | 243,711 | 19,233 | 2.79E+06 | 13,864.8 | ***NATC*** | 1626 | 9000 | 2611 | 2.78E+06 | 228.6 | 65 | 470.0 | **−0.1%** |
| large9 | 16 | 2 | 19 | 21 | 94 | 10 | 11 | 26,006 | 304,140 | 21,820 | 2.85E+06 | 15,097.6 | ***NATC*** | 2155 | 15,600 | 3642 | 2.84E+06 | 551.1 | 49 | 1434.2 | **−0.3%** |

BARON Solver: Best Objective Value: best upper bound; CPU Time (best): time to reach the best upper bound; CPU time (convergence): time to optimal solution.
Proposed heuristics: model size in terms of number of variables and constraints is based on the first iteration of FullP.
NA=Not applicable/not calculable; NATC: Not able to converge.
(Δ) calculated as $\frac{best\ \ solution\ \ heuristics}{best/optimal\ \ solution\ \ BARON} - 1 \rightarrow$ Negative values: better solution from heuristic.

**Table A3**
Computational experiments result with 2 maximum number of disruption event per scenario ($N = 2$).

| dataset | \|K\| | \|OF\| | \|NF\| | \|I\| | \|J\| | \|L\| | \|P\| | Binary Variables | Integer Variables | Constraints | BARON Solver | | | | | | Proposed heuristics | | | | Δ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | Best Objective Value | CPU time (best) | CPU time (convergence) | Binary Variables (first iter) | Integer Variables (first iter) | Constraints (first iter) | Best Objective Value | CPU time (best) | Iteration # (best) | Local Search Total Time | |
| small0 | 3 | 0 | 2 | 2 | 9 | 2 | 3 | 205 | 1755 | 914 | 4.36E+06 | 48.3 | 9776.40 | 94 | 468 | 354 | 4.36E+06 | 0.3 | 17 | 0.4 | – |
| small1 | 4 | 0 | 2 | 2 | 6 | 3 | 2 | 412 | 2142 | 1152 | 5.53E+06 | 229.6 | 18,404.60 | 444 | 1620 | 966 | 5.53E+06 | 0.5 | 5 | 1.8 | – |
| small2 | 2 | 0 | 3 | 3 | 10 | 3 | 2 | 440 | 2340 | 1099 | 4.16E+06 | 194.7 | 5037.00 | 134 | 504 | 374 | 4.16E+06 | 0.2 | 4 | 1.5 | – |
| small3 | 3 | 1 | 2 | 3 | 7 | 2 | 2 | 405 | 2898 | 1405 | 5.23E+06 | 3605.00 | 14,415.40 | 204 | 990 | 647 | 5.23E+06 | 0.2 | 5 | 1.4 | – |
| small4 | 3 | 0 | 3 | 3 | 9 | 2 | 2 | 405 | 3402 | 1530 | 5.57E+06 | 341.9 | 9411.30 | 90 | 468 | 354 | 5.57E+06 | 2.3 | 25 | 3.2 | – |
| small5 | 4 | 1 | 2 | 3 | 8 | 3 | 3 | 808 | 4704 | 2040 | 3.01E+06 | 2058.00 | 15,772.00 | 453 | 1890 | 1057 | 3.01E+06 | 4.4 | 24 | 5.8 | – |
| small6 | 5 | 1 | 2 | 3 | 6 | 3 | 2 | 1019 | 5832 | 2510 | 6.95E+06 | 8223.30 | 13,448.80 | 803 | 4032 | 1958 | 6.95E+06 | 1.1 | 28 | 1.5 | – |
| small7 | 5 | 0 | 3 | 3 | 8 | 2 | 3 | 687 | 6696 | 2722 | 5.72E+06 | 3240.00 | 10,417.70 | 94 | 432 | 336 | 5.72E+06 | 5.4 | 48 | 7.9 | – |
| small8 | 2 | 1 | 5 | 6 | 10 | 2 | 3 | 1350 | 10,152 | 3917 | 5.77E+06 | 4583.10 | 15,281.60 | 94 | 504 | 373 | 5.77E+06 | 0.9 | 1 | 3.7 | – |
| small9 | 5 | 0 | 5 | 5 | 10 | 2 | 2 | 1695 | 18,150 | 5805 | 5.70E+06 | 17,213.40 | 19,676.30 | 30 | 126 | 152 | 5.70E+06 | 0.8 | 5 | 12.9 | – |
| medium0 | 6 | 1 | 7 | 8 | 14 | 5 | 4 | 12,836 | 69,930 | 16,449 | 2.49E+06 | 18,736.00 | ***NATC*** | 1,804 | 7392 | 2893 | 2.49E+06 | 5.2 | 7 | 55.1 | – |
| medium1 | 8 | 0 | 6 | 6 | 18 | 5 | 5 | 9678 | 69,930 | 15,808 | 4.90E+06 | 16,904.00 | ***NATC*** | 3,555 | 18,900 | 5850 | 4.88E+06 | 0.3 | 2 | 44.9 | **−0.5%** |
| medium2 | 7 | 1 | 6 | 7 | 16 | 5 | 6 | 11,312 | 71,190 | 16,129 | 4.90E+06 | 20,872.00 | ***NATC*** | 2,330 | 11,232 | 4035 | 4.51E+06 | 11.4 | 7 | 206.3 | **−7.8%** |
| medium3 | 9 | 1 | 6 | 7 | 22 | 6 | 4 | 17,409 | 123,216 | 24,147 | 3.70E+06 | 40,244.00 | ***NATC*** | 921 | 3780 | 1696 | 3.67E+06 | 255.9 | 64 | 937 | **−0.7%** |
| medium4 | 10 | 2 | 7 | 9 | 16 | 6 | 5 | 31,174 | 193,800 | 35,434 | *Baron runs out of memory* | | | 4,265 | 17,550 | 5587 | 3.76E+06 | 22.7 | 7 | 335.7 | NA |
| medium5 | 9 | 2 | 6 | 8 | 31 | 5 | 5 | 18,654 | 194,157 | 33,124 | *Baron runs out of memory* | | | 1,824 | 13,104 | 4322 | 2.49E+06 | 5.2 | 7 | 1437.2 | NA |
| medium6 | 9 | 1 | 8 | 9 | 25 | 5 | 4 | 23,364 | 211,356 | 35,992 | *Baron runs out of memory* | | | 1,804 | 11,088 | 3817 | 4.52E+06 | 78.9 | 7 | 1295.0 | NA |
| medium7 | 10 | 0 | 8 | 8 | 30 | 7 | 5 | 29,144 | 220,590 | 37,028 | *Baron runs out of memory* | | | 3,232 | 17,280 | 5554 | 3.67E+06 | 255.9 | 64 | 2270.8 | NA |
| medium8 | 9 | 1 | 9 | 10 | 36 | 6 | 5 | 34,623 | 328,320 | 48,549 | *Baron runs out of memory* | | | 3,448 | 23,760 | 6529 | 4.88E+06 | 0.3 | 2 | 873.5 | NA |
| medium9 | 10 | 1 | 9 | 10 | 41 | 7 | 7 | 44,740 | 410,130 | 57,441 | *Baron runs out of memory* | | | 1,972 | 12,852 | 4910 | 4.51E+06 | 11.4 | 7 | 1494.6 | NA |
| large0 | 11 | 1 | 13 | 14 | 61 | 12 | 10 | 165,791 | 1192,425 | 125,019 | *Baron runs out of memory* | | | 15,126 | 86,724 | 18,115 | 4.24E+06 | 397.6 | 7 | 7391.9 | NA |
| large1 | 12 | 3 | 11 | 14 | 85 | 10 | 8 | 148,812 | 1696,383 | 161,304 | *Baron runs out of memory* | | | 4,735 | 41,040 | 11,509 | 3.59E+06 | 958.7 | 7 | 26,837.5 | NA |
| large2 | 12 | 3 | 13 | 16 | 67 | 12 | 8 | 235,740 | 1855,014 | 174,425 | *Baron runs out of memory* | | | 8,705 | 51,975 | 12,503 | 2.94E+06 | 280.2 | 7 | 8069.7 | NA |
| large3 | 14 | 0 | 13 | 13 | 99 | 9 | 11 | 134,222 | 1984,500 | 187,281 | *Baron runs out of memory* | | | 11,400 | 131,868 | 25,620 | 4.06E+06 | 2704.4 | 19 | 13,484.8 | NA |
| large4 | 11 | 2 | 17 | 19 | 64 | 8 | 12 | 214,115 | 2368,710 | 210,859 | *Baron runs out of memory* | | | 7,174 | 62,700 | 14,914 | 2.82E+06 | 124.2 | 8 | 1833.4 | NA |
| large5 | 14 | 1 | 14 | 15 | 96 | 10 | 9 | 197,404 | 2552,580 | 222,059 | *Baron runs out of memory* | | | 21,798 | 139,503 | 25,768 | 2.66E+06 | 45.6 | 7 | 1467.8 | NA |
| large6 | 17 | 1 | 15 | 16 | 59 | 11 | 8 | 297,996 | 2603,601 | 235,863 | *Baron runs out of memory* | | | 12,546 | 128,304 | 25,033 | 4.02E+06 | 5571.5 | 25 | 24,927.2 | NA |
| large7 | 20 | 0 | 18 | 18 | 59 | 10 | 10 | 402,340 | 4092,543 | 335,929 | *Baron runs out of memory* | | | 29,768 | 216,000 | 35,752 | 1.90E+06 | 0.3 | 1 | 238.6 | NA |
| large8 | 19 | 1 | 19 | 20 | 63 | 10 | 8 | 470,009 | 4874,220 | 379,359 | *Baron runs out of memory* | | | 5,826 | 40,500 | 11,536 | 2.83E+06 | 4322.30 | 65 | 10,993.5 | NA |
| large9 | 16 | 2 | 19 | 21 | 94 | 10 | 11 | 445,586 | 5778,660 | 409,432 | *Baron runs out of memory* | | | 7,405 | 70,200 | 16,137 | 2.88E+06 | 8855.70 | 49 | 28,326.2 | NA |

BARON Solver: Best Objective Value: best upper bound; CPU Time (best): time to reach best upper bound; CPU time (convergence): time to optimal solution.
Proposed heuristics: model size in terms of number of variables and constraints is based on first iteration of FullP.
NA=Not applicable/not calculable; NATC: Not able to converge.
(Δ) calculated as $\frac{best\ solution\ heuristics}{best/optimal\ solution\ BARON} - 1 \rightarrow$ Negative values: better solution from heuristic.

## Appendix 3. Real-case application results

**Table A4**
Real-case application results with comparison between Complete Formulation results and No Resilience Actions results.

| Disruption Layers | ϑ | dur | N | Complete Formulation | | | | | | | | | | No Resilience Actions | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Total cost | Investment cost | Operational cost | Primary suppliers | Contracted Capacity | Backup suppliers | Established Facilities | Installed capacity | Protection levels | Recovery actions | Total cost | (Δ) | Investment cost | (Δ) | Operational cost | (Δ) | Primary suppliers | Contracted Capacity | Established Facilities | Installed capacity |
| *Nominal Scenario* | | | | 2.617E+6 | 5.99E+5 | 2.02E+6 | [6, 8] | 4900 | [] | [3, 11] | 4900 | - | – | 2.617E+6 | (-) | 5.985E+5 | (-) | 2.018E+6 | (-) | [6, 8] | 4900 | [3, 11] | 4900 |
| Sup | 0.2 | na | 1 | 2.663E+6 | 6.35E+5 | 2.03E+6 | [6, 8] | 5950 | [] | [3, 11] | 4900 | – | – | 2.663E+6 | (-) | 6.353E+5 | (-) | 2.028E+6 | (-) | [6, 8] | 5950 | [3, 11] | 4900 |
| | | | 2 | 2.666E+6 | 6.48E+5 | 2.02E+6 | [6, 8] | 6300 | [] | [3, 11] | 4900 | – | – | 2.666E+6 | (-) | 6.475E+5 | (-) | 2.018E+6 | (-) | [6, 8] | 6300 | [3, 11] | 4900 |
| | 0.6 | na | 1 | 2.796E+6 | 9.47E+5 | 1.85E+6 | [1, 3, 5, 6] | 5950 | [] | [0, 2, 3, 7] | 5950 | – | – | 2.796E+6 | (-) | 9.468E+5 | (-) | 1.849E+6 | (-) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.861E+6 | 1.02E+6 | 1.84E+6 | [1, 3, 5, 6] | 8050 | [] | [0, 2, 3, 7] | 5950 | – | – | 2.861E+6 | (-) | 1.020E+6 | (-) | 1.841E+6 | (-) | [1, 3, 5, 6] | 8050 | [0, 2, 3, 7] | 5950 |
| | 1 | na | 1 | 2.933E+6 | 1.01E+6 | 1.92E+6 | [1, 3, 5, 6, 8] | 6300 | [10] | [0, 2, 3, 7] | 5600 | – | – | 2.947E+6 | (0.5%) | 9.985E+5 | (−1.%) | 1.948E+6 | (1.2%) | [1, 3, 5, 6, 10] | 6300 | [0, 2, 3, 7] | 5600 |
| | | | 2 | 3.053E+6 | 1.09E+6 | 1.97E+6 | [1, 3, 5, 6, 8, 10] | 7700 | [] | [0, 2, 3, 7] | 5250 | – | – | 3.053E+6 | (-) | 1.087E+6 | (-) | 1.966E+6 | (-) | [1, 3, 5, 6, 8, 10] | 7700 | [0, 2, 3, 7] | 5250 |
| Fac | 0.2 | 2 | 1 | 2.663E+6 | 6.20E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 5600 | – | 1.05E+4 | 2.675E+6 | (0.4%) | 6.300E+5 | (1.7%) | 2.045E+6 | (0.1%) | [6, 8] | 4900 | [3, 11] | 5950 |
| | | | 2 | 2.681E+6 | 6.41E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 6300 | – | – | 2.681E+6 | (-) | 6.405E+5 | (-) | 2.040E+6 | (-) | [6, 8] | 4900 | [3, 11] | 6300 |
| | | 10 | 1 | 2.670E+6 | 6.30E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 5950 | – | 2.10E+3 | 2.675E+6 | (0.2%) | 6.300E+5 | (-) | 2.045E+6 | (0.2%) | [6, 8] | 4900 | [3, 11] | 5950 |
| | | | 2 | 2.681E+6 | 6.41E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 6300 | – | – | 2.681E+6 | (-) | 6.405E+5 | (-) | 2.040E+6 | (-) | [6, 8] | 4900 | [3, 11] | 6300 |
| | | 18 | 1 | 2.672E+6 | 6.30E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 5950 | – | 2.10E+3 | 2.675E+6 | (0.1%) | 6.300E+5 | (-) | 2.045E+6 | (0.1%) | [6, 8] | 4900 | [3, 11] | 5950 |
| | | | 2 | 2.681E+6 | 6.41E+5 | 2.04E+6 | [6, 8] | 4900 | [] | [3, 11] | 6300 | – | – | 2.681E+6 | (-) | 6.405E+5 | (-) | 2.040E+6 | (-) | [6, 8] | 4900 | [3, 11] | 6300 |
| | 0.6 | 2 | 1 | 2.756E+6 | 7.81E+5 | 1.98E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 6300 | – | 6.93E+4 | 2.822E+6 | (2.4%) | 9.468E+5 | (21.3%) | 1.876E+6 | (−5.1%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.817E+6 | 7.88E+5 | 2.03E+6 | [6, 8] | 4900 | [] | [3, 11] | 5600 | 10 | – | 2.909E+6 | (3.2%) | 1.012E+6 | (28.4%) | 1.897E+6 | (−6.5%) | [1, 3, 5, 6] | 6300 | [0, 2, 3, 7] | 7700 |
| | | 10 | 1 | 2.792E+6 | 8.58E+5 | 1.93E+6 | [1, 5, 6] | 4900 | [3] | [0, 2, 3, 7] | 5950 | – | 5.67E+4 | 2.822E+6 | (1.1%) | 9.468E+5 | (10.4%) | 1.876E+6 | (−3.%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.817E+6 | 7.88E+5 | 2.03E+6 | [6, 8] | 4900 | [] | [3, 11] | 5600 | 10 | – | 2.909E+6 | (3.2%) | 1.012E+6 | (28.4%) | 1.897E+6 | (−6.5%) | [1, 3, 5, 6] | 6300 | [0, 2, 3, 7] | 7700 |
| | | 18 | 1 | 2.804E+6 | 8.75E+5 | 1.93E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5250 | 10 | 1.89E+4 | 2.822E+6 | (0.7%) | 9.468E+5 | (8.2%) | 1.876E+6 | (−2.8%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.817E+6 | 7.88E+5 | 2.03E+6 | [6, 8] | 4900 | [] | [3, 11] | 5600 | 10 | – | 2.909E+6 | (3.2%) | 1.012E+6 | (28.4%) | 1.897E+6 | (−6.5%) | [1, 3, 5, 6] | 6300 | [0, 2, 3, 7] | 7700 |
| | 1 | 2 | 1 | 2.819E+6 | 9.00E+5 | 1.92E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5250 | 14 | 2.10E+4 | 3.013E+6 | (6.9%) | 1.082E+6 | (20.2%) | 1.931E+6 | (0.7%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 18] | 6650 |
| | | | 2 | 2.853E+6 | 9.28E+5 | 1.93E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5600 | 15 | 6.30E+4 | 3.221E+6 | (12.9%) | 1.344E+6 | (44.9%) | 1.877E+6 | (−2.5%) | [0, 1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 8, 14, 18] | 7700 |
| | | 10 | 1 | 2.832E+6 | 9.12E+5 | 1.92E+6 | [5, 6, 8] | 4550 | [] | [3, 7, 11] | 5250 | 14 | 2.10E+4 | 3.013E+6 | (6.4%) | 1.082E+6 | (18.6%) | 1.931E+6 | (0.6%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 18] | 6650 |
| | | | 2 | 2.865E+6 | 9.49E+5 | 1.92E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5950 | 15 | 3.15E+4 | 3.221E+6 | (12.4%) | 1.344E+6 | (41.7%) | 1.877E+6 | (−2.1%) | [0, 1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 8, 14, 18] | 7700 |
| | | 18 | 1 | 2.837E+6 | 9.19E+5 | 1.92E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5250 | 15 | 1.05E+4 | 3.013E+6 | (6.2%) | 1.082E+6 | (17.8%) | 1.931E+6 | (0.7%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 18] | 6650 |
| | | | 2 | 2.868E+6 | 9.49E+5 | 1.92E+6 | [5, 6, 8] | 4900 | [] | [3, 7, 11] | 5950 | 15 | 1.05E+4 | 3.221E+6 | (12.3%) | 1.344E+6 | (41.7%) | 1.877E+6 | (−2.2%) | [0, 1, 3, 5, 6] | 6650 | [0, 2, 3, 7, 8, 14, 18] | 7700 |
| Sup+Fac | 0.2 | 2 | 1 | 2.672E+6 | 7.61E+5 | 1.91E+6 | [5, 6, 8] | 5250 | [] | [3, 7, 11] | 5250 | – | 1.47E+4 | 2.678E+6 | (0.2%) | 7.613E+5 | (-) | 1.917E+6 | (0.3%) | [5, 6, 8] | 5250 | [3, 7, 11] | 5250 |
| | | | 2 | 2.708E+6 | 7.96E+5 | 1.91E+6 | [5, 6, 8] | 5950 | [] | [3, 7, 11] | 5600 | – | 1.13E+5 | 2.713E+6 | (0.2%) | 8.068E+5 | (1.3%) | 1.907E+6 | (−0.3%) | [5, 6, 8] | 5950 | [3, 7, 11] | 5950 |
| | | 10 | 1 | 2.674E+6 | 7.61E+5 | 1.91E+6 | [5, 6, 8] | 5250 | [] | [3, 7, 11] | 5250 | – | 1.47E+4 | 2.678E+6 | (0.1%) | 7.613E+5 | (-) | 1.917E+6 | (0.2%) | [5, 6, 8] | 5250 | [3, 7, 11] | 5250 |
| | | | 2 | 2.713E+6 | 8.07E+5 | 1.91E+6 | [5, 6, 8] | 5950 | [] | [3, 7, 11] | 5950 | – | 1.68E+4 | 2.713E+6 | (0.%) | 8.068E+5 | (-) | 1.907E+6 | (0.%) | [5, 6, 8] | 5950 | [3, 7, 11] | 5950 |
| | | 18 | 1 | 2.676E+6 | 7.61E+5 | 1.91E+6 | [5, 6, 8] | 5250 | [] | [3, 7, 11] | 5250 | – | 8.40E+3 | 2.678E+6 | (0.1%) | 7.613E+5 | (-) | 1.917E+6 | (0.1%) | [5, 6, 8] | 5250 | [3, 7, 11] | 5250 |
| | | | 2 | 2.713E+6 | 8.07E+5 | 1.91E+6 | [5, 6, 8] | 5950 | [] | [3, 7, 11] | 5950 | – | 6.30E+3 | 2.713E+6 | (0.%) | 8.068E+5 | (-) | 1.907E+6 | (0.%) | [5, 6, 8] | 5950 | [3, 7, 11] | 5950 |
| | 0.6 | 2 | 1 | 2.789E+6 | 9.47E+5 | 1.84E+6 | [1, 3, 5, 6] | 5950 | [] | [0, 2, 3, 7] | 5950 | – | 5.67E+4 | 2.809E+6 | (0.7%) | 9.468E+5 | (-) | 1.863E+6 | (1.1%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.879E+6 | 1.02E+6 | 1.86E+6 | [1, 3, 5, 6] | 7000 | [] | [0, 2, 3, 7] | 7000 | – | 4.16E+5 | 2.912E+6 | (1.1%) | 1.015E+6 | (-) | 1.897E+6 | (1.8%) | [1, 3, 5, 6] | 7000 | [0, 2, 3, 7] | 7000 |
| | | 10 | 1 | 2.796E+6 | 9.47E+5 | 1.85E+6 | [1, 3, 5, 6] | 5950 | [] | [0, 2, 3, 7] | 5950 | – | 4.41E+4 | 2.809E+6 | (0.5%) | 9.468E+5 | (-) | 1.863E+6 | (0.7%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.890E+6 | 1.02E+6 | 1.87E+6 | [1, 3, 5, 6] | 7000 | [] | [0, 2, 3, 7] | 7000 | – | 3.28E+5 | 2.912E+6 | (0.8%) | 1.015E+6 | (-) | 1.897E+6 | (1.2%) | [1, 3, 5, 6] | 7000 | [0, 2, 3, 7] | 7000 |
| | | 18 | 1 | 2.802E+6 | 9.47E+5 | 1.86E+6 | [1, 3, 5, 6] | 5950 | [] | [0, 2, 3, 7] | 5950 | – | 4.41E+4 | 2.809E+6 | (0.3%) | 9.468E+5 | (-) | 1.863E+6 | (0.4%) | [1, 3, 5, 6] | 5950 | [0, 2, 3, 7] | 5950 |
| | | | 2 | 2.900E+6 | 1.02E+6 | 1.88E+6 | [1, 3, 5, 6] | 7000 | [] | [0, 2, 3, 7] | 7000 | – | 3.09E+5 | 2.912E+6 | (0.4%) | 1.015E+6 | (-) | 1.897E+6 | (0.7%) | [1, 3, 5, 6] | 7000 | [0, 2, 3, 7] | 7000 |
| | 1 | 2 | 1 | 2.928E+6 | 9.92E+5 | 1.94E+6 | [1, 3, 5, 6] | 6650 | [] | [0, 2, 3, 7] | 6650 | – | 1.37E+5 | 3.010E+6 | (2.8%) | 9.923E+5 | (-) | 2.017E+6 | (4.2%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7] | 6650 |
| | | | 2 | 3.116E+6 | 1.12E+6 | 2.00E+6 | [1, 3, 5, 6] | 5950 | [8, 10] | [0, 2, 3, 7] | 5950 | 17 | 2.10E+5 | 3.294E+6 | (5.7%) | 1.246E+6 | (11.2%) | 2.048E+6 | (2.6%) | [1, 3, 5, 6, 10] | 8400 | [0, 2, 3, 7, 18] | 8400 |
| | | 10 | 1 | 2.953E+6 | 9.92E+5 | 1.96E+6 | [1, 3, 5, 6] | 6650 | [] | [0, 2, 3, 7] | 6650 | – | 1.16E+5 | 3.010E+6 | (1.9%) | 9.923E+5 | (-) | 2.017E+6 | (2.9%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7] | 6650 |
| | | | 2 | 3.120E+6 | 1.12E+6 | 2.00E+6 | [1, 3, 5, 6] | 5950 | [8, 10] | [0, 2, 3, 7] | 5950 | 17 | 1.79E+5 | 3.294E+6 | (5.6%) | 1.246E+6 | (11.2%) | 2.048E+6 | (2.4%) | [1, 3, 5, 6, 10] | 8400 | [0, 2, 3, 7, 18] | 8400 |
| | | 18 | 1 | 2.974E+6 | 1.00E+6 | 1.97E+6 | [1, 3, 5, 6] | 6300 | [8] | [0, 2, 3, 7] | 6300 | 2 | 8.40E+4 | 3.010E+6 | (1.2%) | 9.923E+5 | (−0.8%) | 2.017E+6 | (2.2%) | [1, 3, 5, 6] | 6650 | [0, 2, 3, 7] | 6650 |
| | | | 2 | 3.122E+6 | 1.11E+6 | 2.02E+6 | [1, 3, 5, 6] | 5950 | [8, 10] | [0, 2, 3, 7] | 5600 | 19 | 7.35E+4 | 3.294E+6 | (5.5%) | 1.246E+6 | (12.7%) | 2.048E+6 | (1.6%) | [1, 3, 5, 6, 10] | 8400 | [0, 2, 3, 7, 18] | 8400 |

## Appendix 4. Robustness vs Resilience design comparison

**Table A5**
Selected design used for the comparison in the Robustness vs Resilience section.

| | Nominal design | | Resilient design #1 | | Robust design #1 | | Resilient design #2 | | Robust design #2 | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Site | Capacity | Site | Capacity | Site | Capacity | Site | Capacity | Site | Capacity |
| *Primary suppliers* | Sup#6 | 2800 | Sup#5 | 1400 | Sup#5 | 1400 | Sup#1 | 1400 | Sup#1 | 1750 |
| | Sup#8 | 2100 | Sup#6 | 2100 | Sup#6 | 2100 | Sup#3 | 1400 | Sup#3 | 1750 |
| | | | Sup#8 | 2450 | Sup#8 | 2450 | Sup#5 | 1050 | Sup#5 | 1400 |
| | | | | | | | Sup#6 | 1750 | Sup#6 | 1750 |
| | | | | | | | | | Sup#10 | 1750 |
| | *total* | *4900* | *total* | *5950* | *total* | *5950* | *total* | *5950* | *total* | *8400* |
| *Backup suppliers* | – | – | n/a | Sup#8 | n/a | | | | | |
| | | | | Sup#10 | | | | | | |
| *Facilities* | Fac#3 | 2800 | Fac#3 | 1750 | Fac#3 | 2100 | Fac#0 | 1400 | Fac#0 | 1750 |
| | Fac#11 | 2100 | Fac#7 | 1750 | Fac#7 | 1400 | Fac#3 | 1400 | Fac#2 | 1750 |
| | | | Fac#11 | 2100 | Fac#11 | 2450 | Fac#7 | 1750 | Fac#3 | 1750 |
| | | | | | | | Fac#11 | 1050 | Fac#7 | 1400 |
| | | | | | | | | | Fac#18 | 1750 |
| | *total* | *4900* | *total* | *5600* | *total* | *5950* | *total* | *5600* | *total* | *8400* |
| *Protection systems* | – | Fac#3 | – | – | Fac#0 | 100% | – | | | |
| | | Fac#7 | – | | Fac#3 | 100% | | | | |
| | | Fac#11 | – | | Fac#7 | 100% | | | | |
| | | | | | Fac#11 | 80% | | | | |

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